The derivative and the indefinite integral

A workbook to be written in, covering lessons 1 to 48. Each topic runs Focus (easy), Practice (medium), Challenge (hard) and Beyond (very hard, olympiad). Answers, with solutions for the Beyond questions, are at the back.

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17 topics · 765 questions · 45 per topic

Increments, and the problem of the tangent

The step across, the step up, and the ratio between them — the one quotient the whole of this term is built on.

Lessons 1–2 · Algebra and Calculus 11, §1.1 · Pure Mathematics 1, 7.1 · 45 questions

KEY WORDS

- increment of the argument
- increment of the function
- difference quotient
- average rate of change
- secant
- tangent
- gradient
- limiting position
- instantaneous rate

REMEMBER

1. Increment of the argument $\Delta x = x_2 - x_1$; increment of the function $\Delta y = f(x_1 + \Delta x) - f(x_1)$.
2. The quotient $\frac{\Delta y}{\Delta x}$ is the gradient of the secant through the two points.
3. Average rate of change on $[a, b]$ is $\frac{f(b) - f(a)}{b - a}$.
4. The tangent is the limiting position of the secant as $\Delta x \to 0$.
5. Simplify $\frac{\Delta y}{\Delta x}$ completely before letting $\Delta x \to 0$ — the h in the denominator must cancel.

Focus

EASY · 10 questions

The basic skill, one step at a time.

1. $f(x) = 3x + 1$. Find Δy as x goes from 2 to 5.

2. $f(x) = x^2$. Find Δy as x goes from 1 to 4.

3. $f(x) = x^2$. Find $\frac{\Delta y}{\Delta x}$ on $[1, 4]$.

4. $f(x) = 2x - 7$. Find Δy when $\Delta x = 0.5$.

5. $f(x) = x^2$. Find Δy at $x = 3$ with $\Delta x = h$.

6 $f(x) = x^2$. Simplify $\frac{\Delta y}{\Delta x}$ at $x = 3$.

7 $f(x) = 5$. Find Δy for any Δx.

8 Find the gradient of the secant to $y = x^2$ through $(1, 1)$ and $(3, 9)$.

9 Find the average rate of change of $f(x) = x^3$ on $[0, 2]$.

10 A car covers 120 km in 2 hours. Find its average speed.

Practice

MEDIUM 10 questions

The same skill with more to simplify first.

1 $f(x) = x^2 + 2x$. Find Δy at $x = 1$ with $\Delta x = h$.

2 For $f(x) = x^2 + 2x$ simplify $\frac{\Delta y}{\Delta x}$ at $x = 1$.

3 $f(x) = \frac{1}{x}$. Find Δy at $x = 2$ with $\Delta x = h$.

4 For $f(x) = \frac{1}{x}$ simplify $\frac{\Delta y}{\Delta x}$ at $x = 2$.

5 $f(x) = x^3$. Find Δy at $x = 1$ with $\Delta x = h$.

6 For $f(x) = x^3$ simplify $\frac{\Delta y}{\Delta x}$ at $x = 1$.

7 Find the gradient of the secant to $y = \frac{1}{x}$ through $x = 1$ and $x = 2$.

8 Find the average rate of change of $f(x) = x^2 - 3x$ on $[2, 5]$.

9 $s(t) = t^2 + t$. Find the average speed on $[1, 3]$.

10 $f(x) = \sqrt{x}$. Find Δy at $x = 4$ with $\Delta x = h$.

Challenge

HARD 10 questions

Letters in place of numbers, and more than one idea at once.

1 $f(x) = x^2$. Show that the secant through $x = a$ and $x = a + h$ has gradient $2a + h$.

2 $f(x) = \frac{1}{x^2}$. Simplify $\frac{\Delta y}{\Delta x}$ at $x = 1$.

3 $f(x) = \sqrt{x}$. Simplify $\frac{\Delta y}{\Delta x}$ at $x = a$.

4 $f(x) = ax^2 + bx + c$. Simplify $\frac{\Delta y}{\Delta x}$ at x.

5 $f(x) = x^2$. Find h so that the secant through $x = 2$ and $x = 2 + h$ has gradient 5.

6 $f(x) = \frac{1}{x}$. Find the gradient of the secant through $x = a$ and $x = b$.

7 A stone falls $s = 5t^2$. Find its average speed on $[1, 1 + h]$.

8 State the value $10 + 5h$ approaches as $h \to 0$, and what it measures.

9 $f(x) = x^3$. Simplify $\frac{\Delta y}{\Delta x}$ at $x = a$.

10 $f(x) = \frac{x}{x + 1}$. Find Δy at $x = 0$ with $\Delta x = h$.

Beyond VERY HARD 15 questions

Olympiad work. Expect to spend real time on these — some need an idea, not just a method.

1. $f(x) = x^n$, n a positive integer. Show that the secant gradient through $x = a$ and $x = a + h$ tends to na^{n-1} as $h \to 0$.

2. $f(x) = x^2$. Find the point of $[1, 3]$ at which the tangent gradient equals the secant gradient over $[1, 3]$.

3. Prove that for any quadratic the secant gradient over $[a, b]$ equals the tangent gradient at the midpoint.

4. $f(x) = \dfrac{1}{x}$ with $0 < a < b$. Find the point of (a, b) at which the tangent gradient equals the secant gradient.

5. $f(x) = |x|$. Find the average rate of change on $[-1, 1]$, and explain why the curve has no tangent at $x = 0$.

6 Show that the secant to $y = x^3$ over $[-a, a]$ has gradient a^2.

7 $f(x) = x^2$. The secant through $x = p$ and $x = q$ has gradient $p + q$. Use this to find the tangent gradient at $x = p$.

8 Find every a for which the secant to $y = x^2$ over $[a, a + 2]$ has gradient 0.

9 A point moves with $s = t^3 - 3t$. Find the average velocity on $[1, 1 + h]$ and its limit as $h \to 0$. What does the limit say?

10 Show that the secant to $y = \sqrt{x}$ over $[a, b]$ has gradient $\frac{1}{\sqrt{a} + \sqrt{b}}$.

11 Two points of $y = x^2$ have x-coordinates t and $t + 2$. Show that the secant gradient equals the tangent gradient at $x = t + 1$.

12 $f(x) = x^2 + bx$. Find b so that the secant over $[0, 4]$ has gradient 1.

13 Prove that for $f(x) = \dfrac{1}{x}$ the secant gradient is negative for every pair $a, b > 0$ with $a \neq b$.

14 The average rate of change of f over $[0, x]$ is $x + 1$ for every $x > 0$, and $f(0) = 0$. Find $f(x)$.

15 $f(x) = x^2$. Find the limit of $\dfrac{f(2 + h) - f(2 - h)}{2h}$ as $h \to 0$, and compare it with the ordinary quotient.

The limit of a function

What a function does near a point, not at it — and the four techniques that get past the form $\frac{0}{0}$.

Lessons 3–4 · Algebra and Calculus 11, §1.2 · Pure Mathematics 1, 7.2 · 45 questions

KEY WORDS

- limit
- tends to
- one-sided limit
- indeterminate form
- conjugate expression
- continuous function
- point of discontinuity
- removable discontinuity
- infinity

REMEMBER

1. $\lim_{x \to a} f(x) = L$ describes f near a. The value $f(a)$ itself is irrelevant, and need not exist.
2. The limit exists only when the two one-sided limits agree.
3. Substitute first. If it gives a number, that number is the limit.
4. The form $\frac{0}{0}$ means a common factor is hiding: factorise and cancel, or multiply by the conjugate.
5. For $x \to \infty$ in a rational function, divide every term by the highest power of x.

Focus

EASY · 10 questions

The basic skill, one step at a time.

1. Find $\lim_{x \to 2} (3x + 1)$.

2. Find $\lim_{x \to 0} (x^2 - 4)$.

3. Find $\lim_{x \to 3} \frac{x^2 - 9}{x - 3}$.

4. Find $\lim_{x \to 1} \frac{x^2 - 1}{x - 1}$.

5. Find $\lim_{x \to -2} \frac{x + 2}{x^2 - 4}$.

6 Find $\lim_{x \to 4} \sqrt{x}$.

7 Find $\lim_{x \to 0} \frac{5x}{x}$.

8 Find $\lim_{x \to \infty} \frac{1}{x}$.

9 Find $\lim_{x \to 2} \frac{x^3 - 8}{x - 2}$.

10 State the value of $\lim_{x \to 0} \frac{\sin x}{x}$, with x in radians.

Practice

MEDIUM 10 questions

The same skill with more to simplify first.

1 Find $\lim_{x \to 3} \frac{x^2 - 5x + 6}{x - 3}$.

2 Find $\lim_{x \to 0} \frac{\sqrt{x + 9} - 3}{x}$.

3 Find $\lim_{x \to \infty} \frac{3x^2 + 2x}{x^2 - 1}$.

4 Find $\lim_{x \to \infty} \frac{2x + 1}{x^2 + 3}$.

5 Find $\lim_{x\to\infty} \frac{x^3+1}{x^2+1}$.

6 Find $\lim_{x\to 0} \frac{\tan x}{x}$.

7 Find $\lim_{x\to 1} \frac{x^3-1}{x^2-1}$.

8 Find $\lim_{x\to 0} \frac{1}{x}(\frac{1}{1+x} - 1)$.

9 Find $\lim_{x\to 2} \frac{x^2-4}{x^2-3x+2}$.

10 Find $\lim_{x\to 0} \frac{\sin 3x}{x}$.

Challenge

HARD 10 questions

Letters in place of numbers, and more than one idea at once.

1 Find $\lim_{x\to 0} \frac{\sin 5x}{\sin 2x}$.

2 Find $\lim_{x\to 0} \frac{1-\cos x}{x^2}$.

3 Find $\lim_{x\to\infty} (\sqrt{x^2+1}-x)$.

4 Find $\lim_{x\to 1} \frac{\sqrt{x}-1}{x-1}$.

5 Find $\lim_{x\to 0} \frac{\sqrt{1+x}-\sqrt{1-x}}{x}$.

6 Find $\lim_{x\to 2} \frac{x^3-3x-2}{x^2-4}$.

7 State the value of $\lim_{x\to\infty} (1+\frac{1}{x})^x$.

8 Find $\lim_{x\to 0} \frac{e^x-1}{x}$.

9 $f(x) = \dfrac{x^2 - 4}{x - 2}$ for $x \neq 2$, and $f(2) = a$. Find a so that f is continuous at 2.

10 Find the two one-sided limits of $\dfrac{|x|}{x}$ at $x = 0$, and say whether the limit exists.

Beyond

VERY HARD · 15 questions

Olympiad work. Expect to spend real time on these — some need an idea, not just a method.

1 Find $\lim_{x \to 0} \dfrac{(1 + x)^{1/3} - 1}{x}$.

2 Find $\lim_{x \to 1} \dfrac{x^n - 1}{x - 1}$ for a positive integer n.

3 Find $\lim_{x \to 0} \dfrac{\sin ax}{\sin bx}$ for constants a and $b \neq 0$.

4 Find $\lim_{x\to\infty}(\sqrt{x^2+x}-x)$.

5 Find $\lim_{x\to 0}\dfrac{\tan x-\sin x}{x^3}$.

6 Find $\lim_{x\to 0}\dfrac{1-\cos 2x}{x\cdot\sin x}$.

7 Find $\lim_{x\to 2}\dfrac{\sqrt{x+2}-2}{x-2}$.

8 State the value of $\lim_{x\to 0}(1+x)^{1/x}$ and explain its link to $\lim_{x\to\infty}(1+\dfrac{1}{x})^{x}$.

9 Use the squeeze theorem to show that $\lim_{x\to 0} x \cdot \sin \frac{1}{x} = 0$.

10 Find $\lim_{x\to \infty} x \cdot \sin \frac{1}{x}$.

11 Find a and b so that $\lim_{x\to \infty} (\frac{x^2+1}{x+1} - ax - b) = 0$.

12 Find $\lim_{x\to 1} (\frac{1}{1-x} - \frac{3}{1-x^3})$.

13 Find $\lim_{x\to 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{(1+x)^{1/3} - (1-x)^{1/3}}$.

14 Show that $\lim_{x \to 0} \frac{\sin x}{x} = 1$ fails when x is measured in degrees, and find the correct value.

15 $f(x) = \frac{x^2 - 1}{|x| - 1}$. Find the limits of f as $x \to 1$ and as $x \to -1$.

The derivative of a function

The limit of the difference quotient — differentiation from first principles, and what the answer measures.

Lessons 5–6 · Algebra and Calculus 11, §1.3 · Pure Mathematics 1, 7.1–7.2 · 45 questions

KEY WORDS

- derivative
- differentiation
- from first principles
- gradient function
- differentiable
- instantaneous velocity
- Leibniz notation
- corner point
- rate of change

REMEMBER

1. $f'(x) = \lim_{h \to 0} \dfrac{f(x + h) - f(x)}{h}$.
2. Four steps: write $f(x + h)$; form Δy and simplify; divide by h; let $h \to 0$.
3. $f'(a)$ is the gradient of the tangent at $x = a$, and the instantaneous rate of change there.
4. $f'(x)$, y' and $\dfrac{dy}{dx}$ all name the same thing.
5. Differentiable at a point ⇒ continuous there. The converse is false: $|x|$ at $x = 0$.

Focus

EASY · 10 questions

The basic skill, one step at a time.

1. $f(x) = 7$. Find $f'(x)$ from first principles.

2. $f(x) = 4x$. Find $f'(x)$ from first principles.

3. $f(x) = 3x - 2$. Find $f'(x)$.

4. $f(x) = x^2$. Find $f'(x)$ from first principles.

5. $f(x) = x^2$. Find $f'(5)$.

6. $f(x) = x^3$. Find $f'(x)$ from first principles.

7 $f(x) = x^2 + 1$. Find $f'(x)$.

8 Give the gradient of the tangent to $y = x^2$ at $(3, 9)$.

9 $s(t) = t^2$. Find the velocity at $t = 4$.

10 Write the derivative of $y = x^2$ in Leibniz notation.

Practice

MEDIUM 10 questions

The same skill with more to simplify first.

1 $f(x) = 2x^2 - 3x$. Find $f'(x)$ from first principles.

2 $f(x) = \frac{1}{x}$. Find $f'(x)$ from first principles.

3 $f(x) = \frac{1}{x}$. Find $f'(2)$.

4 $f(x) = \sqrt{x}$. Find $f'(x)$ from first principles.

5 $f(x) = \sqrt{x}$. Find $f'(9)$.

6 $f(x) = x^2 - 4x + 1$. Find where $f'(x) = 0$.

7 $f(x) = 5x - x^2$. Find $f'(3)$.

8 $s(t) = 3t^2 + 2t$. Find the velocity at $t = 1$.

9 $f(x) = x^3 - x$. Find $f'(2)$.

10 $f(x) = \frac{1}{x^2}$. Find $f'(x)$ from first principles.

Challenge

HARD 10 questions

Letters in place of numbers, and more than one idea at once.

1 $f(x) = \frac{1}{x + 1}$. Find $f'(x)$ from first principles.

2 $f(x) = \sqrt{x + 3}$. Find $f'(x)$ from first principles.

3 $f(x) = ax^2 + bx + c$. Find $f'(x)$ from first principles.

4 $f(x) = x^4$. Find $f'(x)$ from first principles.

5 Show that $f(x) = |x|$ has no derivative at $x = 0$.

6 $f(x) = x|x|$. Find $f'(0)$.

7 $f(x) = \frac{1}{\sqrt{x}}$. Find $f'(x)$ from first principles.

8 $s(t) = t^3 - 6t^2$. Find every instant at which the velocity is zero.

9 $f(x) = x^2$. Find the point at which the tangent gradient is 7.

10 $f(x) = x^3$. Find every point at which the tangent gradient is 12.

Beyond

VERY HARD · 15 questions

Olympiad work. Expect to spend real time on these — some need an idea, not just a method.

1 $f(x) = x^n$ for a positive integer n. Find $f'(x)$ from first principles.

2 $f(x) = x^{1/3}$, the cube root of x. Find $f'(x)$ from first principles.

3 $f(x) = \sin x$. Find $f'(x)$ from first principles.

4 $f(x) = \cos x$. Find $f'(x)$ from first principles.

5 Prove that a function differentiable at a is continuous at a.

6 $f(x) = x^2$ for $x \leq 1$ and $f(x) = 2x - 1$ for $x > 1$. Is f differentiable at $x = 1$?

7 $f(x) = x^2$ for $x \leq 1$ and $f(x) = 3x - 2$ for $x > 1$. Is f differentiable at $x = 1$?

8 $f(x) = x^2 \cdot \sin\frac{1}{x}$ for $x \neq 0$ and $f(0) = 0$. Find $f'(0)$.

9 Find a and b so that $f(x) = x^2$ for $x \leq 2$ and $f(x) = ax + b$ for $x > 2$ is differentiable at $x = 2$.

10 The tangent to $y = x^2$ at $x = a$ passes through $(0, -1)$. Find a.

11 Show that the only horizontal tangent to $y = x^3$ is at the origin.

12 $f(x) = x|x|$. Show that $f'(x) = 2|x|$ for every x, including $x = 0$.

13 A particle has $s = t^3 - 3t^2 + 2$. Find its velocity, and every instant at which it is at rest.

14 Show that $f(x) = x^{2/3}$ has no derivative at $x = 0$.

15 Show that the tangent to $y = x^2$ at $x = a$ meets the curve at no other point.

The rules of differentiation

The table of derivatives, and the four rules that differentiate anything built out of it.

Lessons 7–9 · Algebra and Calculus 11, §1.4 · Pure Mathematics 1, 7.3 · 45 questions

KEY WORDS

- power rule
- constant multiple rule
- sum rule
- product rule
- quotient rule
- fractional index
- negative index
- simplify first
- second derivative

REMEMBER

1. $(x^n)' = nx^{n-1}$, for every real n — whole, negative or fractional.
2. $(cf)' = cf'$ and $(u \pm v)' = u' \pm v'$: constants come out, sums split up.
3. Product rule: $(uv)' = u'v + uv'$.
4. Quotient rule: $\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}$ — the order in the numerator matters.
5. Rewrite roots and reciprocals as powers first. There is no product or quotient rule you need if the expression simplifies.

Focus

EASY · 10 questions

The basic skill, one step at a time.

1. Differentiate $y = x^5$.

2. Differentiate $y = 3x^4$.

3. Differentiate $y = x^2 + x$.

4. Differentiate $y = 7$.

5. Differentiate $y = x^{-3}$.

6. Differentiate $y = \sqrt{x}$.

7 Differentiate $y = \frac{1}{x}$.

8 Differentiate $y = 4x^3 - 2x^2 + 5$.

9 Differentiate $y = x^2 \cdot x^3$.

10 Find $\frac{dy}{dx}$ at $x = 1$ for $y = x^3$.

Practice

MEDIUM 10 questions

The same skill with more to simplify first.

1 Differentiate $y = (x + 1)(x + 2)$.

2 Differentiate $y = x^2(x - 3)$.

3 Differentiate $y = \frac{x^2 + 1}{x}$.

4 Differentiate $y = x\sqrt{x}$.

5 Differentiate $y = \frac{3}{x^2}$.

6 Differentiate $y = (2x - 1)(x + 4)$.

7 Differentiate $y = \frac{x^3 - 1}{x^2}$.

8 Differentiate $y = 5\sqrt{x} - \frac{2}{x}$.

9 Differentiate $y = (x + 1)^2$.

10 Find the second derivative of $y = x^4$.

Challenge

HARD 10 questions

Letters in place of numbers, and more than one idea at once.

1 Differentiate $y = (x^2 + 1)(x^3 - 2)$.

2 Differentiate $y = \frac{x}{x + 1}$.

3 Differentiate $y = \frac{x - 1}{x + 1}$.

4 Differentiate $y = \frac{x^2 + 1}{x - 1}$.

5 Differentiate $y = x^3\sqrt{x}$.

6 Differentiate $y = (\sqrt{x} + 1)(\sqrt{x} - 1)$.

7 Differentiate $y = \frac{2x + 1}{\sqrt{x}}$.

8 Differentiate $y = \frac{x^2}{x^2 + 1}$.

9 Find $f''(x)$ for $f(x) = \frac{1}{x}$.

10 Find the gradient of $y = \frac{x + 2}{x - 2}$ at $x = 4$.

Beyond

VERY HARD · 15 questions

Olympiad work. Expect to spend real time on these — some need an idea, not just a method.

1. Prove the product rule $(uv)' = u'v + uv'$ from first principles.

2. Deduce the quotient rule from the product rule.

3. Prove that $(uvw)' = u'vw + uv'w + uvw'$.

4. Show that $\left(\frac{1}{v}\right)' = -\frac{v'}{v^2}$.

5. Differentiate $y = (x - 1)(x - 2)(x - 3)$.

6 Find every x at which $y = x^3 - 3x$ has gradient 0.

7 $f(x) = x^2 g(x)$, where $g(2) = 3$ and $g'(2) = -1$. Find $f'(2)$.

8 $f(x) = \dfrac{x^2 + a}{x + 1}$ has a stationary point at $x = 1$. Find a.

9 Show that $y = x + \dfrac{1}{x}$ has gradient zero exactly at $x = 1$ and $x = -1$.

10 Find the second derivative of $y = \dfrac{x + 1}{x - 1}$.

11 Prove that the derivative of an even function is odd.

12 Prove that the derivative of an odd function is even.

13 $f(x) = x^n$ for a positive integer n. Show that the n-th derivative is the constant $n!$.

14 Find every polynomial f with $f'(x) = f(x)$ for all x.

15 $y = x^3 + ax + b$ has a stationary point at $x = 2$ and passes through $(0, 1)$. Find a and b.

The derivative of a composite function

The chain rule — differentiate the outside, then multiply by the derivative of the inside.

Lessons 10–12 · Algebra and Calculus 11, §1.5 · Pure Mathematics 1, 7.4 · Pure Mathematics 2, 4.4 · 45 questions

KEY WORDS

- chain rule
- composite function
- inner function
- outer function
- intermediate variable
- bracket power
- substitution u
- connected rates

REMEMBER

1. $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$ — the intermediate variable cancels like a fraction.
2. $[f(g(x))]' = f'(g(x)) \cdot g'(x)$: outer derivative at the inner value, times the inner derivative.
3. $[(ax + b)^n]' = na(ax + b)^{n-1}$. The factor a is the mark most often lost.
4. Name the inner function before you write anything else.
5. Chains combine with the product and quotient rules: deal with the outer structure first.

Focus EASY 10 questions — The basic skill, one step at a time.

1. Differentiate $y = (x + 1)^5$.

2. Differentiate $y = (2x + 3)^4$.

3. Differentiate $y = (3x - 1)^2$.

4. Differentiate $y = (x^2 + 1)^3$.

5. Differentiate $y = \sqrt{x + 4}$.

6. Differentiate $y = (5x)^3$.

7 Differentiate $y = (1 - x)^6$.

8 Differentiate $y = (x^3 + 2)^2$.

9 Differentiate $y = \dfrac{1}{x + 1}$.

10 Differentiate $y = (4x - 7)^{10}$.

Practice MEDIUM 10 questions

The same skill with more to simplify first.

1 Differentiate $y = \sqrt{x^2 + 1}$.

2 Differentiate $y = (x^2 - 3x)^4$.

3 Differentiate $y = \dfrac{1}{(2x + 1)^3}$.

4 Differentiate $y = (1 + \sqrt{x})^3$.

5 Differentiate $y = \sqrt{3x - 2}$.

6 Differentiate $y = (x + \dfrac{1}{x})^2$.

7 Differentiate $y = (2 - x^2)^5$.

8 Differentiate $y = \frac{1}{\sqrt{x + 1}}$.

9 Find $\frac{dy}{dx}$ at $x = 0$ for $y = \sqrt{x^2 + 1}$.

10 Differentiate $y = (\frac{x + 1}{x - 1})^2$.

Challenge

HARD 10 questions

Letters in place of numbers, and more than one idea at once.

1 Differentiate $y = x^2(2x + 1)^3$.

2 Differentiate $y = x\sqrt{x + 1}$.

3 Differentiate $y = \frac{(x + 1)^3}{x - 1}$.

4 Differentiate $y = \sqrt{x^2 + x + 1}$.

5 Differentiate $y = (1 + \sqrt{1 + x})^2$.

6 Differentiate $y = (\frac{2x - 1}{x + 2})^3$.

7 Find $\frac{dy}{dx}$ at $x = 1$ for $y = (x^2 + 3)^4$.

8 Find $\frac{dy}{dx}$ at $x = 0$ for $y = \sqrt{1 - x^2}$.

9 Differentiate $y = (3x + 1)^{1/3}$.

10 The radius of a circle grows at 2 cm/s. Find $\frac{dA}{dt}$ when $r = 5$ cm.

Beyond VERY HARD 15 questions

Olympiad work. Expect to spend real time on these — some need an idea, not just a method.

1 Explain why $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$, using increments.

2 Differentiate $y = ((x^2 + 1)^3 + 1)^2$.

3 Differentiate $y = \sqrt{x + \sqrt{x}}$.

4 $y = f(x^2)$ and $f'(1) = 3$. Find $\frac{dy}{dx}$ at $x = 1$.

5 $y = (f(x))^2$, with $f(2) = 3$ and $f'(2) = -1$. Find $\frac{dy}{dx}$ at $x = 2$.

6 A spherical balloon is inflated at 100 cm³/s. Find the rate at which the radius grows when $r = 5$ cm.

7 $y = (ax + b)^n$. Show that the n-th derivative is the constant $n!a^n$.

8 Differentiate $y = \dfrac{x}{\sqrt{1 + x^2}}$ and simplify.

9 A ladder 5 m long leans against a wall. Its foot slides out at 0.5 m/s. Find the rate at which the top slides down when the foot is 3 m from the wall.

10 $y = (1 + x)^n$. Show that $\dfrac{dy}{dx}$ at $x = 0$ is n, and say how that matches the binomial expansion.

11 $g(x) = f(2x + 1)$ and $f'(5) = 4$. Find $g'(2)$.

12 Differentiate $y = \sqrt{1 + \sqrt{1 + x}}$.

13 Show that $y = \sqrt{x^2 + 1}$ satisfies $y \cdot \frac{dy}{dx} = x$.

14 The edge of a cube grows at 3 cm/s. Find the rate of change of its surface area when the edge is 10 cm.

15 $y = f(g(x))$, where $f'(t) > 0$ for every t and $g'(x) < 0$ for every x. Show that y is decreasing.

Revision · Control work 1 — the derivative

Everything from lessons 1 to 12 in one place: limits, first principles, the rules, and the chain rule.

Lessons 13–14 · Control work 1 and work on the mistakes · 45 questions

KEY WORDS

- limit
- first principles
- power rule
- product rule
- quotient rule
- chain rule
- second derivative
- gradient
- rate of change

REMEMBER

1. Substitute first. The form $\frac{0}{0}$ means factorise, or use the conjugate.
2. Use first principles only when the question says so; otherwise use the rules.
3. Simplify before differentiating — it is nearly always quicker than the quotient rule.
4. The chain rule needs the inner derivative. Leaving it out is the commonest lost mark in this unit.
5. Check a derivative by substituting a value, or by differentiating a simplified form of the same function.

Focus

EASY · 10 questions

The basic skill, one step at a time.

1. Find $\lim_{x \to 1} (x^2 + 3)$.

2. Find $\lim_{x \to 2} \frac{x^2 - 4}{x - 2}$.

3. Differentiate $y = x^7$.

4. Differentiate $y = 5x^3 - 2x$.

5. Differentiate $y = \frac{1}{x^3}$.

6 Find $\frac{dy}{dx}$ at $x = 4$ for $y = \sqrt{x}$.

7 Differentiate $y = (x + 2)^4$.

8 $f(x) = 3x^2$. Find $f'(x)$ from first principles.

9 Differentiate $y = x(x + 1)$.

10 Find the second derivative of $y = x^3$.

Practice

MEDIUM 10 questions

The same skill with more to simplify first.

1 Find $\lim_{x \to 0} \frac{\sqrt{x + 4} - 2}{x}$.

2 Find $\lim_{x \to \infty} \frac{2x^2 + 1}{3x^2 - x}$.

3 Differentiate $y = (2x - 1)(x + 3)$.

4 Differentiate $y = \frac{x}{x^2 + 1}$.

5 Differentiate $y = \sqrt{2x + 1}$.

6 Differentiate $y = (x^2 - 1)^3$.

7 $f(x) = \frac{2}{x}$. Find $f'(x)$ from first principles.

8 Differentiate $y = x^2 + \frac{1}{x}$.

9 Find the gradient of $y = x^3 - x$ at $x = 2$.

10 Find where $y = x^2 - 6x$ has gradient 0.

Challenge

HARD 10 questions

Letters in place of numbers, and more than one idea at once.

1 Find $\lim_{x \to 0} \frac{\sin 4x}{3x}$.

2 Differentiate $y = x^2\sqrt{x + 1}$.

3 Differentiate $y = \frac{x - 1}{x^2 + 1}$.

4 Differentiate $y = \frac{1 + x}{1 - x}$.

5 $f(x) = \sqrt{2x}$. Find $f'(x)$ from first principles.

6 Find $\frac{dy}{dx}$ at $x = 0$ for $y = (3x^2 + 1)^5$.

7 Find a if $y = x^3 + ax$ has gradient 5 at $x = 1$.

8 Differentiate $y = \frac{1}{(x^2 + 1)^2}$.

9 Find the second derivative of $y = \sqrt{x}$.

10 Find $\lim_{x \to 1} \frac{x^3 - 1}{x - 1}$.

Beyond VERY HARD 15 questions

Olympiad work. Expect to spend real time on these — some need an idea, not just a method.

1 Find $\lim_{x\to 0} \dfrac{\sqrt{1+2x}-\sqrt{1-2x}}{x}$.

2 Differentiate $y = \dfrac{x}{1+\sqrt{x}}$ and simplify.

3 $f(x) = x^3 - 3x^2 + 2$. Find f', f'', and the point at which $f'' = 0$.

4 Show that $(x^n)' = nx^{n-1}$ still holds when n is a negative integer.

5 $f(x)g(x) = x^2$ for all x, with $f(1) = 1$, $f'(1) = 2$ and $g(1) = 1$. Find $g'(1)$.

6 Show that $\lim_{x \to 0} \dfrac{1 - \cos 3x}{x^2} = \dfrac{9}{2}$.

7 $y = (x^2 + a)^3$. Find every value of a for which the curve has exactly one point of zero gradient.

8 Find every x at which $y = x^3 - 3x^2 + 3x$ has gradient 0.

9 A particle has $s = \sqrt{t^2 + 1}$. Find its velocity at $t = 2$.

10 $z = f(2x)$. Express $\dfrac{dz}{dx}$ in terms of f', and find $\dfrac{d^2z}{dx^2}$.

11 Verify Rolle's theorem for $f(x) = x^2 - 4x + 3$ on $[1, 3]$.

12 Find $\lim_{x \to 2} \dfrac{\sqrt{x} - \sqrt{2}}{x - 2}$.

13 Show that $(fg)'' = f''g + 2f'g' + fg''$.

14 Find the cubic f with $f(0) = 1$, $f'(0) = 0$, $f''(0) = 2$ and $f'''(0) = 6$.

15 $y = (1 + x)^{100}$. Find $y'(0)$ and $y''(0)$.

The modulus function

Distance read off the number line — the graph of $y = |ax + b|$, and why the corner has no gradient.

Lessons 15–16 · Pure Mathematics 2, 1.1–1.2 · Cambridge insert · 45 questions

KEY WORDS

- modulus
- absolute value
- distance from zero
- vertex
- corner point
- piecewise definition
- one-sided derivative
- critical point
- case method

REMEMBER

1. $|x| = x$ for $x \geq 0$ and $-x$ for $x < 0$. Also $|x| = \sqrt{x^2}$ — never just x.
2. $|x - a|$ is the distance from x to a. This reading solves most questions faster than algebra.
3. $|f(x)| = k$ gives $f(x) = k$ or $f(x) = -k$ when $k > 0$, and nothing at all when $k < 0$.
4. $|x| < k$ means $-k < x < k$; $|x| > k$ means $x < -k$ or $x > k$ — two pieces, joined by “or”.
5. $y = |f(x)|$ reflects the part below the axis upwards; $y = f(|x|)$ mirrors the right-hand part into the left.

Focus

EASY · 10 questions

The basic skill, one step at a time.

1. Evaluate $|-7|$.

2. Evaluate $|3 - 8|$.

3. Solve $|x| = 4$.

4. Solve $|x - 2| = 3$.

5. Solve $|x| = -2$.

6. Solve $|2x| = 10$.

7 Evaluate $\sqrt{(-5)^2}$.

8 Write the distance from x to 7 as a modulus.

9 Solve $|x + 1| = 0$.

10 State the vertex of $y = |x - 3|$.

Practice

MEDIUM 10 questions

The same skill with more to simplify first.

1 Solve $|2x - 1| = 7$.

2 Solve $|3x + 2| = 5$.

3 Solve $|x| < 3$.

4 Solve $|x| > 5$.

5 Solve $|x - 4| \leq 2$.

6 Solve $|x + 3| > 1$.

7 State the vertex of $y = |2x + 6|$.

8 State the vertex of $y = |x| - 2$.

9 Solve $|x - 1| = |x + 3|$.

10 State the gradient of $y = |x|$ for $x > 0$.

Challenge

HARD 10 questions

Letters in place of numbers, and more than one idea at once.

1 Solve $|x - 2| + |x + 1| = 5$.

2 Solve $|2x - 1| = x + 2$.

3 Solve $|x^2 - 4| = 0$.

4 Solve $|x|^2 - 3|x| + 2 = 0$.

5 Solve $|x - 1| < 2x$.

6 Find the two one-sided derivatives of $y = |x|$ at $x = 0$.

7 Show that $\frac{d}{dx}|x| = \frac{x}{|x|}$ for $x \neq 0$.

8 Solve $|x + 2| = |2x - 1|$.

9 State the x-coordinates of the corners of $y = |x^2 - 1|$.

10 Solve $|x| + x = 6$.

Beyond

VERY HARD · 15 questions

Olympiad work. Expect to spend real time on these — some need an idea, not just a method.

1 Solve $|x - 1| + |x - 3| = 2$.

2 Solve $|x - 1| + |x - 3| = 1$.

3 Solve $|x - 1| + |x - 3| = 6$.

4 Prove the triangle inequality $|a + b| \leq |a| + |b|$.

5 Prove that $||a| - |b|| \leq |a - b|$.

6 $f(x) = x|x|$. Show that f is differentiable everywhere, but that $f''(0)$ does not exist.

7 Solve $|x^2 - 5x + 4| = x^2 - 5x + 4$.

8 Find the least value of $f(x) = |x - 1| + |x - 2| + |x - 3|$.

9 Describe the graph of $y = |x| + |x - 2|$, and say where it is constant.

10 Solve $|x - 1| = x - 1$.

11 Solve $|x - 1| + |x + 1| \le 4$.

12 Explain why $\sqrt{x^2} = |x|$ and not x.

13 Find every a for which the graph of $y = |x - a|$ passes through $(1, 3)$.

14 Show that $|x|$ is continuous at $x = 0$ but not differentiable there.

15 Solve $|x - 2| < |x + 1|$.

The equations of the tangent and the normal

One point and one gradient fix a line — and the normal is its perpendicular at the same point.

Lessons 17–18 · Algebra and Calculus 11, §1.6 · Pure Mathematics 1, 7.8 · 45 questions

KEY WORDS

- tangent
- normal
- point of tangency
- gradient of a tangent
- perpendicular gradients
- point-gradient form
- horizontal tangent
- intercept

REMEMBER

1. Tangent at $x = a$: $y - f(a) = f'(a)(x - a)$.
2. Normal gradient is $-\dfrac{1}{f'(a)}$, through the same point.
3. Find $f(a)$ and $f'(a)$ before writing a line down. Two numbers, then one formula.
4. If $f'(a) = 0$ the tangent is horizontal, $y = f(a)$, and the normal is vertical, $x = a$.
5. Perpendicular gradients multiply to -1.

Focus

EASY · 10 questions

The basic skill, one step at a time.

1. $y = x^2$. Find the tangent gradient at $x = 2$.

2. $y = x^2$. Find the tangent at $(2, 4)$.

3. $y = x^2$. Find the normal gradient at $x = 2$.

4. $y = x^3$. Find the tangent gradient at $x = 1$.

5. $y = 3x + 1$. Find the tangent at $(0, 1)$.

6. $y = x^2$. Find where the tangent is horizontal.

7 $y = x^2 - 4x$. Find the tangent gradient at $x = 1$.

8 A line has gradient 2. Find the gradient of a line perpendicular to it.

9 $y = \frac{1}{x}$. Find the tangent gradient at $x = 1$.

10 $y = x^2$. Find the normal at $(0, 0)$.

Practice

MEDIUM 10 questions

The same skill with more to simplify first.

1 $y = x^2 + 1$. Find the tangent at $(1, 2)$.

2 $y = x^3 - x$. Find the tangent at $(1, 0)$.

3 $y = \sqrt{x}$. Find the tangent at $(4, 2)$.

4 $y = \frac{1}{x}$. Find the tangent at $(1, 1)$.

5 $y = x^2$. Find the normal at $(1, 1)$.

6 $y = x^2 - 2x$. Find the normal at $(2, 0)$.

7 $y = x^3$. Find every point at which the tangent gradient is 3.

8 $y = x^2 + 3x$. Find where the tangent is horizontal.

9 $y = 2x^2$. Find the tangent at $(1, 2)$.

10 Find the y-intercept of the tangent to $y = x^2$ at $(3, 9)$.

Challenge

HARD 10 questions

Letters in place of numbers, and more than one idea at once.

1 $y = x^2$. The tangent at $x = a$ passes through $(0, -4)$. Find a.

2 $y = \frac{1}{x}$. Find the tangent at $(2, \frac{1}{2})$.

3 $y = x^3 - 3x$. Find the tangents parallel to the x-axis.

4 $y = \sqrt{x}$. Find the normal at $(1, 1)$.

5 $y = x^2$. Find the point at which the normal has gradient 1.

6 $y = x^2 - 4$. The tangent at $x = 1$ meets the x-axis. Find where.

7 $y = \dfrac{x + 1}{x - 1}$. Find the tangent gradient at $x = 2$.

8 $y = x^3$. The tangent at $(1, 1)$ meets the curve again. Find that point.

9 Find the area of the triangle the tangent to $y = x^2$ at $(2, 4)$ cuts from the axes.

10 $y = x^2 + bx$. The tangent at $x = 0$ has gradient 3. Find b.

Beyond

VERY HARD · 15 questions

Olympiad work. Expect to spend real time on these — some need an idea, not just a method.

1 Find the equations of both tangents to $y = x^2$ that pass through $(0, -1)$.

2 $y = x^2$. Find where the normal at $(1, 1)$ meets the curve again.

3 The tangent to $y = \frac{1}{x}$ at $(a, \frac{1}{a})$, with $a > 0$, cuts the axes at A and B. Show that the area of triangle OAB does not depend on a.

4 Show that the point of tangency is the midpoint of the part of the tangent cut off by the axes, for $y = \frac{1}{x}$.

5 Find c for which the line $y = x + c$ is a tangent to $y = x^2$.

6 Find every m for which $y = mx - 1$ is a tangent to $y = x^2$.

7 The tangent and the normal to $y = x^2$ at $(1, 1)$ meet the y-axis at P and Q. Find the length PQ.

8 Show that the tangent to $y = x^3$ at $x = a$ meets the curve again at $x = -2a$.

9 Find the points of $y = x^3 - 3x$ at which the tangent is parallel to $y = 9x$.

10 The curves $y = x^2$ and $y = \sqrt{x}$ meet at $(1, 1)$. Find the angle between them there.

11 Do $y = x^2$ and $y = \frac{1}{x}$ cut at right angles where they meet?

12 Find k for which $y = x^2 + k$ touches the line $y = 2x$.

13 The normal to $y = x^2$ at (a, a^2), with $a \neq 0$, meets the y-axis at $(0, b)$. Find b in terms of a.

14 Show that the normal to $y = x^2$ at (a, a^2), with $a \neq 0$, meets the curve again at $x = -a - \frac{1}{2a}$.

15 Find every point of $y = x^3$ at which the tangent passes through $(0, 2)$.

Investigating a function with the derivative

Where a curve rises, where it falls, where it turns — and the sign chart that settles all three.

Lessons 19–22 · Algebra and Calculus 11, §1.7 · Pure Mathematics 1, 8.1–8.2 · 45 questions

KEY WORDS

- increasing function
- decreasing function
- stationary point
- critical point
- local maximum
- local minimum
- point of inflection
- sign chart
- curve sketching

REMEMBER

1. $f' > 0$ on an interval means increasing there; $f' < 0$ means decreasing.
2. Stationary points are the solutions of $f'(x) = 0$.
3. First derivative test: + then − is a maximum, − then + a minimum, no change an inflection.
4. Second derivative test at a stationary point: $f'' < 0$ maximum, $f'' > 0$ minimum, $f'' = 0$ tells you nothing — go back to the sign chart.
5. $f'' > 0$ is concave up and $f'' < 0$ concave down; an inflection needs f'' to change sign, not merely to vanish.

Focus

EASY · 10 questions

The basic skill, one step at a time.

1. $y = x^2 - 4x$. Find the stationary point.

2. $y = x^2 - 4x$. State whether it is a maximum or a minimum.

3. $y = x^3 - 3x$. Find $\frac{dy}{dx}$.

4. $y = x^3 - 3x$. Find the stationary points.

5. $y = x^2$. State where it is increasing.

6. $y = 4 - x^2$. State where it is decreasing.

7 $y = x^2 + 2x$. Find $\frac{d^2y}{dx^2}$.

8 $y = x^3$. Find $\frac{d^2y}{dx^2}$.

9 $y = x^3$. Find where $\frac{d^2y}{dx^2} = 0$.

10 State the nature of the stationary point of $y = 5 - x^2$.

Practice

MEDIUM 10 questions

The same skill with more to simplify first.

1 $y = x^3 - 3x^2$. Find the stationary points.

2 $y = x^3 - 3x^2$. Classify each stationary point.

3 $y = x^3 - 3x^2$. Find the point of inflection.

4 $y = 2x^3 - 9x^2 + 12x$. Find the stationary points.

5 State where $y = x^3 - 12x$ is increasing.

6 $y = x^4 - 2x^2$. Find the stationary points.

7 $y = x + \frac{1}{x}$. Find the stationary points.

8 $y = x + \frac{1}{x}$. State the nature of the stationary point at $x = 1$.

9 $y = x^2(x - 3)$. Find $\frac{dy}{dx}$.

10 Find the greatest value of $y = 6x - x^2$.

Challenge

HARD 10 questions

Letters in place of numbers, and more than one idea at once.

1 $y = x^3 - 6x^2 + 9x$. Find and classify every stationary point.

2 $y = x^3 - 6x^2 + 9x$. Find the point of inflection.

3 $y = \frac{x}{x^2 + 1}$. Find the stationary points.

4 $y = x^4 - 4x^3$. Find the stationary points.

5 $y = x^4 - 4x^3$. State the nature of the stationary point at $x = 0$.

6 Find the greatest and least values of $y = x^3 - 3x$ on $[0, 2]$.

7 $y = \dfrac{x - 1}{x + 1}$. State whether it is increasing or decreasing.

8 Show that $y = x^3 + x$ is increasing for every x.

9 $y = x^4 + 4x$. Find and classify the stationary point.

10 Find every a for which $y = x^3 + ax$ has no stationary point.

Beyond

VERY HARD · 15 questions

Olympiad work. Expect to spend real time on these — some need an idea, not just a method.

1 $y = x^3 + ax^2 + bx$ has stationary points at $x = 1$ and $x = 3$. Find a and b.

2 Show that a cubic has either two stationary points or none, apart from the borderline case of one repeated.

3 $y = x^4 - 4x^3 + 10$. Find and classify every stationary point.

4 Prove that a differentiable function is monotonic between two consecutive stationary points.

5 Find the least value of $y = x + \frac{1}{x}$ for $x > 0$.

6 Deduce from the previous question that $x + \frac{1}{x} \geq 2$ for every $x > 0$, and state when equality holds.

7 Show that $y = ax^3 + bx^2 + cx + d$ has its point of inflection at $x = -\frac{b}{3a}$.

8 Find the greatest value of $y = x^2(3 - x)$ on $[0, 3]$.

9 Find the greatest and least values of $y = \frac{x}{x^2 + 1}$ over all real x.

10 Show that $f(x) = x^3 - 3x^2 + 3x - 1$ has exactly one stationary point and no maximum or minimum.

11 $y = x^3 + px + q$ has a stationary point lying on the x-axis. Find the relation between p and q.

12 Show that if $f''(x) > 0$ for every x, then f has at most one stationary point.

13 $y = \dfrac{x^2 + 1}{x}$. Find the stationary points and their nature.

14 Find the number of real roots of $x^3 - 3x + k = 0$, in terms of k.

15 Prove that a polynomial of degree n has at most $n - 1$ stationary points.

Extremum problems

Name the quantity, write the constraint, eliminate, differentiate — and then justify.

Lessons 23–25 · Algebra and Calculus 11, §1.8 · Pure Mathematics 1, 8.3 · 45 questions

KEY WORDS

- extremum
- greatest value
- least value
- constraint
- objective function
- feasible domain
- endpoint value
- optimisation
- justification

REMEMBER

1. Write the quantity to be made greatest or least, then the constraint relating the variables.
2. Eliminate until one variable is left, and state its feasible domain — lengths are positive.
3. Solve $f' = 0$, then justify the nature with f'' or a sign chart. An unjustified answer is an unearned one.
4. On a closed interval, compare the critical values with the value at each endpoint.
5. Answer the question actually asked, with units. The value of x is rarely what was wanted.

Focus

EASY · 10 questions

The basic skill, one step at a time.

1. Two numbers add to 10. Find their greatest product.

2. Find the greatest value of $y = x(10 - x)$.

3. A rectangle has perimeter 20 cm. Find its greatest area.

4. Find the least value of $x^2 + 4$.

5. Find the greatest value of $9 - x^2$.

6. Find the least value of $x^2 - 6x$.

7. Find x for which $x + x^2$ is least.

8. Find the least value of $(x - 2)^2 + 3$.

9. Find the greatest value of $y = 8x - x^2$.

10. Find the least value of $y = x^2 + 2x + 5$.

Practice

MEDIUM · 10 questions · The same skill with more to simplify first.

1. Two numbers add to 12. Find the least possible sum of their squares.

2. A rectangle has perimeter 36 cm. Find its greatest area.

3. Two positive numbers have product 36. Find their least sum.

4. A rectangle has area 64 cm². Find its least perimeter.

5. Find the least value of $x + \dfrac{9}{x}$ for $x > 0$.

6. Find the greatest value of $y = x(6 - x)$ on $[0, 4]$.

7 Find the least value of $y = x^2 + \frac{16}{x}$ for $x > 0$.

8 A box has a square base of edge x and volume 32 cm³. Find x for least total surface area.

9 A farmer has 100 m of fence for a rectangular pen against a straight wall, using the wall as one side. Find the greatest area.

10 Find the least value of $y = (x - 1)^2 + (x - 3)^2$.

Challenge

HARD 10 questions

Letters in place of numbers, and more than one idea at once.

1 Squares of side x are cut from the corners of a 12 cm square sheet and the sides folded up. Find the greatest volume.

2 Find the greatest area of a rectangle inscribed in a semicircle of radius r, with one side on the diameter.

3 Find the greatest area of a rectangle with its base on the x-axis and its top corners on $y = 4 - x^2$.

4 A closed cylinder has volume 1000 cm³. Find the radius giving least total surface area.

5 Find the point of $y = x^2$ nearest to $(0, 3)$.

6 A wire of length 100 cm is cut in two; one piece forms a square and the other a circle. Find the length of the square piece for least total area.

7 Find the least value of $f(x) = x^2 + \frac{1}{x^2}$ for $x > 0$.

8 A rectangular field beside a straight river is fenced on three sides with 240 m of fence. Find the greatest area.

9 Find the greatest value of $y = x(a - x)^2$ on $[0, a]$, for $a > 0$.

10 Find the point of the line $y = 2x + 3$ nearest to the origin.

Beyond VERY HARD 15 questions

Olympiad work. Expect to spend real time on these — some need an idea, not just a method.

1. Find the greatest volume of a cone inscribed in a sphere of radius R.

2. Find the greatest volume of a cylinder inscribed in a sphere of radius R.

3. An open cylinder — no lid — has volume V. Show that the least surface area occurs when the height equals the radius.

4. Two corridors of widths a and b meet at a right angle. Find the length of the longest ladder that can be carried round the corner, kept horizontal.

5. Light travels at speed v_1 in one medium and v_2 in another. Show that the path of least time obeys $\dfrac{\sin\theta_1}{v_1} = \dfrac{\sin\theta_2}{v_2}$.

6 Find the greatest area of a rectangle inscribed in a right-angled triangle with legs a and b, the rectangle sharing the right angle.

7 Find the point of $y = x^2$ nearest to $(0, 2)$, and the shortest distance.

8 Find the greatest value of $f(x) = x^2(1 - x)^3$ on $[0, 1]$.

9 An open box with a square base has volume V. Find the base edge giving least surface area.

10 Two sides of a triangle have fixed lengths a and b. Find the greatest possible area.

11 Use calculus to prove that $\frac{a + b}{2} \geq \sqrt{ab}$ for positive a and b.

12 A rectangular beam is cut from a circular log of diameter d. Its strength is proportional to the width times the square of the depth. Find the strongest beam.

13 Prove that among all rectangles of a given perimeter the square has the greatest area.

14 Find the dimensions of the cylinder of greatest volume that can be inscribed in a cone of base radius R and height H.

15 A window is a rectangle surmounted by a semicircle. Its perimeter is 10 m. Find the greatest possible area.

Revision · Control work 2 — applications of the derivative

Tangents, monotonicity, turning points and optimisation together — the whole of Unit 2, and the quarter.

Lessons 26–27 · Control work 2 and the quarter review · 45 questions

KEY WORDS

- tangent
- normal
- stationary point
- maximum
- minimum
- point of inflection
- optimisation
- sign chart
- justification

REMEMBER

1. A tangent or a normal needs $f(a)$ and $f'(a)$ first — two numbers, then one formula.
2. Stationary points come from $f' = 0$; classify them with f'' or a sign chart.
3. On a closed interval, the greatest value may sit at an endpoint rather than at a turning point.
4. In an optimisation problem, state the feasible domain and justify the extremum.
5. Check the answer for sense. A negative length, or a maximum below an endpoint value, means a slip.

Focus

EASY · 10 questions

The basic skill, one step at a time.

1. $y = x^2$. Find the tangent at $(1, 1)$.

2. $y = x^2 - 2x$. Find the stationary point.

3. $y = x^2 - 2x$. State its nature.

4. $y = x^3 - 12x$. Find $\frac{dy}{dx}$.

5. Find the greatest value of $y = 10 - x^2$.

6 $y = x^3$. Find the normal gradient at $x = 1$.

7 State where $y = x^2$ is decreasing.

8 Two numbers add to 8. Find their greatest product.

9 $y = x^2 + 1$. Find $\frac{d^2y}{dx^2}$.

10 Find the least value of $y = x^2 + 6x + 10$.

Practice

MEDIUM 10 questions

The same skill with more to simplify first.

1 $y = x^3 - 6x^2 + 9x$. Find the stationary points.

2 $y = x^3 - 6x^2 + 9x$. Classify them.

3 $y = 2x^3 - 9x^2$. Find the point of inflection.

4 $y = \frac{2}{x}$. Find the tangent at $(1, 2)$.

5 A rectangle has perimeter 40 cm. Find its greatest area.

6 Find the greatest value of $y = x(4 - x)$.

7 $y = x^4 - 4x^2$. Find the stationary points.

8 $y = \sqrt{x}$. Find the normal at $(4, 2)$.

9 State where $y = x^3 - 3x^2$ is increasing.

10 Find the least value of $y = x + \frac{4}{x}$ for $x > 0$.

Challenge

HARD 10 questions

Letters in place of numbers, and more than one idea at once.

1 $y = x^3 - 3x^2 + 2$. Find and classify every stationary point.

2 Find the greatest and least values of $y = x^3 - 3x$ on $[-2, 3]$.

3 The tangent to $y = x^2$ at (a, a^2) passes through $(1, 0)$. Find a.

4 Squares of side x are cut from the corners of an 18 cm square sheet and the sides folded up. Find the greatest volume.

5 $y = \dfrac{x + 1}{x - 2}$. Find the tangent gradient at $x = 3$.

6 Find a so that $y = x^3 + ax^2$ has a point of inflection at $x = 1$.

7 A closed cylinder has volume 500 cm³. Find the radius giving least surface area.

8 Show that $y = \dfrac{x}{x + 1}$ is increasing on each of its branches.

9 Find the greatest area of a rectangle inscribed in a semicircle of radius 4, with one side on the diameter.

10 Find the greatest value of $y = x^2(6 - x)$ on $[0, 6]$.

Beyond VERY HARD 15 questions

Olympiad work. Expect to spend real time on these — some need an idea, not just a method.

1 $y = x^3 + ax^2 + bx + c$ has a maximum at $x = -1$, a minimum at $x = 3$, and passes through $(0, 5)$. Find a, b and c.

2 Find the number of real roots of $x^3 - 3x = k$, in terms of k.

3 Find the shortest distance from the origin to the line $y = x + 2$.

4 The tangent and the normal to $y = x^2$ at (a, a^2), with $a > 0$, meet the y-axis. Find the area of the triangle they form with it.

5 Find the greatest area of a rectangle inscribed in the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, with sides parallel to the axes.

6 Prove that of all rectangles with a given diagonal, the square has the greatest area.

7 Show that $f(x) = x^3 + px + q$ is increasing for every x exactly when $p \geq 0$.

8 $y = x^4 - 8x^2$. Find and classify every stationary point.

9 A particle moves with $s = t^3 - 6t^2 + 9t$. Find the total distance travelled in the first 4 seconds.

10 Prove that the point of inflection of a cubic is the mean of its two stationary points, whenever those exist.

11 Find the greatest area of a triangle inscribed in a circle of radius R.

12 Show that $f(x) = x + \sin x$ is increasing for every x.

13 $y = ax^3 + bx^2$ has a stationary point at $(2, 4)$. Find a and b.

14 Show that among all cylinders of a given total surface area, the one of greatest volume has its height equal to its diameter.

15 Show that the tangents to $y = x^2$ at $x = a$ and $x = b$, with $a \neq b$, meet at the point $(\frac{a + b}{2}, ab)$.

Approximate calculations with the derivative

The tangent standing in for the curve — good enough for a square root, and only over a short step.

Lessons 28–30 · Algebra and Calculus 11, §1.9 · Pure Mathematics 1, 8.4 · Pure Mathematics 2, 6 · 45 questions

KEY WORDS

- approximation
- linear approximation
- small increment
- differential
- absolute error
- relative error
- percentage error
- convenient point
- order of accuracy

REMEMBER

1. $\Delta y \approx f'(x) \cdot \Delta x$, so $f(x + \Delta x) \approx f(x) + f'(x) \cdot \Delta x$.
2. Choose a convenient point — one where f and f' are exact, and close to the value wanted.
3. $dy = f'(x)dx$ is the rise along the tangent; Δy is the rise along the curve. They differ by an amount of order $(\Delta x)^2$.
4. Relative error is $\left|\frac{\Delta y}{y}\right|$; percentage error is 100 times it.
5. A p % error in a length gives about $2p$ % in an area and $3p$ % in a volume.

Focus

EASY · 10 questions

The basic skill, one step at a time.

1. $f(x) = x^2$. Estimate Δy at $x = 3$ with $\Delta x = 0.01$.

2. $f(x) = x^3$. Estimate Δy at $x = 2$ with $\Delta x = 0.1$.

3. $f(x) = \sqrt{x}$. Estimate Δy at $x = 9$ with $\Delta x = 0.1$.

4. Estimate $\sqrt{101}$.

5. Estimate $\sqrt{4.02}$.

6. Estimate $(2.01)^2$.

7. Estimate $(1.002)^3$.

8. $f(x) = \dfrac{1}{x}$. Estimate Δy at $x = 2$ with $\Delta x = 0.1$.

9. Estimate $\dfrac{1}{1.01}$.

10. Write down the small-increment formula.

Practice

MEDIUM · 10 questions · The same skill with more to simplify first.

1. Estimate $\sqrt{26}$.

2. Estimate $(8.1)^{1/3}$.

3. Estimate $(3.02)^3$.

4. Estimate $\dfrac{1}{\sqrt{4.04}}$.

5. A square has side 10 cm measured to within 0.1 cm. Estimate the error in its area.

6. For that square, find the percentage error in the area.

7 A circle has radius 5 cm with an error of 0.05 cm. Estimate the error in its area.

8 A cube has edge 6 cm with an error of 0.02 cm. Estimate the error in its volume.

9 Estimate $\sin 0.02$, with the angle in radians.

10 Estimate $\tan 0.01$, with the angle in radians.

Challenge

HARD 10 questions

Letters in place of numbers, and more than one idea at once.

1 Estimate $\sqrt{102}$ to three decimal places.

2 Estimate $(1001)^{1/3}$ to four decimal places.

3 Estimate $\sin 31°$ to three decimal places.

4 Estimate $\ln 1.05$.

5 Estimate $e^{0.02}$.

6 A sphere has radius 10 cm with an error of 0.1 cm. Find the percentage error in its volume.

7 The radius of a circle increases by 2 %. Find the percentage increase in its area.

8 Estimate $(2.01)^4$.

9 Show that $(1 + x)^n \approx 1 + nx$ for small x.

10 Estimate $\sqrt{0.98}$.

Beyond VERY HARD 15 questions

Olympiad work. Expect to spend real time on these — some need an idea, not just a method.

1 Show that the error in the linear approximation of f at a is of order $(\Delta x)^2$.

2 Estimate $\sqrt{50}$, and say how far the estimate is from the true value.

3 Use the tangent at $x = 0$ to show $\sqrt{1 + x} \approx 1 + \frac{x}{2}$, and test it on $\sqrt{1.1}$.

4 Compare the linear estimate of $(1.1)^{10}$ with the true value, and say what the comparison shows.

5 The period of a pendulum is $T = 2\pi\sqrt{\frac{L}{g}}$. If L increases by 1 %, find the percentage change in T.

6 A cone has fixed height. If its base radius increases by 3 %, find the percentage increase in its volume.

7 Show that for $f(x) = \frac{1}{x}$ with $x > 0$ the linear approximation always underestimates.

8 Use the iteration $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ with $f(x) = x^2 - 2$ and $x_0 = 1.5$ to find $\sqrt{2}$ to four decimal places.

9 Use the same iteration to solve $x^3 - x - 1 = 0$, starting from $x_0 = 1.5$, to four decimal places.

10 Show that the relative error in x^n is n times the relative error in x.

11 Estimate $\cos 59°$ to four decimal places.

12 The series for $\sqrt{1 + x}$ begins $1 + \frac{x}{2} - \frac{x^2}{8}$. Use the extra term to improve the estimate of $\sqrt{1.02}$.

13 A quantity x is measured with a 1 % error. Find the percentage error in $\frac{1}{x^2}$.

14 Show that the linear estimate of $f(a + h)$ is exact when f is linear.

15 Find how large h may be so that the linear estimate of $(1 + h)^3$ is within 0.01 of the true value.

Modelling with the derivative

A rate in the world, written as a derivative — motion, growth, cost — then read back as an answer with units.

Lessons 31–34 · Algebra and Calculus 11, §1.10 · Pure Mathematics 1, 8.4 · 45 questions

KEY WORDS

- mathematical model
- displacement
- velocity
- acceleration
- connected rates
- marginal cost
- marginal revenue
- units of a rate
- instant of rest

REMEMBER

1. $v = \frac{ds}{dt}$ and $a = \frac{dv}{dt} = \frac{d^2s}{dt^2}$. At rest means $v = 0$, not $s = 0$.
2. Connected rates run through the chain rule: $\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$. Substitute the numbers last.
3. Displacement is $s(t_2) - s(t_1)$; distance travelled splits the journey at every instant of rest.
4. Marginal cost is $C'(x)$ and marginal revenue $R'(x)$; profit is greatest where they are equal.
5. Every answer carries units, and a negative rate means the quantity is falling.

Focus

EASY · 10 questions

The basic skill, one step at a time.

1. $s = t^2 + 3t$. Find v.

2. $s = t^2 + 3t$. Find v at $t = 2$.

3. $s = t^3$. Find the acceleration.

4. $v = 3t^2 - 12t$. Find every instant at which the particle is at rest.

5. $s = 5t^2$. Find v at $t = 1$.

6 s is in metres and t in seconds. State the units of $\frac{ds}{dt}$.

7 $C(x) = 100 + 5x$. Find the marginal cost.

8 $R(x) = 20x - x^2$. Find the marginal revenue.

9 $s = 10t - t^2$. Find the greatest displacement.

10 $v = 4t$. Find the acceleration.

Practice

MEDIUM 10 questions

The same skill with more to simplify first.

1 $s = t^3 - 6t^2 + 9t$. Find v and a.

2 For that motion, find every instant at which the particle is at rest.

3 For that motion, find the acceleration at $t = 2$.

4 A balloon's radius grows at 0.5 cm/s. Find $\frac{dV}{dt}$ when $r = 10$ cm.

5 Water is poured into a cylinder of radius 5 cm at 50 cm³/s. Find the rate the level rises.

6 $P(x) = 50x - x^2 - 200$. Find the output giving greatest profit.

7 $C(x) = x^2 + 10x + 50$. Find the marginal cost at $x = 5$.

8 The side of a square grows at 2 cm/s. Find $\frac{dA}{dt}$ when the side is 6 cm.

9 $s = \sqrt{t + 1}$. Find v at $t = 3$.

10 $v = 6t - t^2$. Find the greatest velocity.

Challenge

HARD 10 questions

Letters in place of numbers, and more than one idea at once.

1 $s = t^3 - 3t^2 - 9t + 5$, with $t \geq 0$. Find when the particle is at rest.

2 $s = t^2 - 4t$. Find the total distance travelled in the first 4 seconds.

3 A ladder 13 m long leans against a wall; its foot slides out at 2 m/s. Find the rate the top falls when the foot is 5 m out.

4 A conical tank has radius half its depth. Water drains at 8 cm³/s. Find the rate the depth falls when the depth is 4 cm.

5 A man 2 m tall walks at 1.5 m/s away from a lamp 6 m high. Find the rate his shadow lengthens.

6 A circular oil slick grows in area at 4 m²/s. Find $\frac{dr}{dt}$ when $r = 10$ m.

7 $R = 100x - 2x^2$ and $C = 20x + 100$. Find the output giving greatest profit.

8 $s = 2t^3 - 15t^2 + 24t$. Find the least velocity on $0 \le t \le 5$.

9 A snowball melts so that its volume falls at a rate proportional to its surface area. Show that the radius falls at a constant rate.

10 $v = t^2 - 5t + 6$. Find the distance travelled in the first 4 seconds.

Beyond VERY HARD 15 questions

Olympiad work. Expect to spend real time on these — some need an idea, not just a method.

1. $s = t^3 - 9t^2 + 24t$. Find the total distance travelled in the first 5 seconds.

2. Water flows at 10 cm³/s into a cone, apex down, whose radius equals its depth. Find the rate the depth rises when it is 5 cm.

3. A searchlight 100 m from a straight shore turns at 0.1 rad/s. Find the speed of the spot of light when the beam makes $30°$ with the perpendicular.

4. Show that profit is greatest where marginal revenue equals marginal cost.

5. Demand is $p = 100 - 2x$. Find the output giving greatest revenue, and the price there.

6 A kite flies 100 m above the ground and moves horizontally at 8 m/s. Find the rate the string pays out when 200 m is out.

7 Show that if $s = at^2 + bt + c$ the acceleration is constant, and identify it.

8 A particle has $a = 6t$, with $v(0) = 1$ and $s(0) = 2$. Find $s(t)$.

9 The side of an equilateral triangle grows at 2 cm/s. Find the rate its area grows when the side is 10 cm.

10 Two cars leave a crossroads at the same moment, one north at 60 km/h and one east at 80 km/h. Find the rate at which the distance between them grows.

11 The surface area of a cube grows at 12 cm²/s. Find the rate its volume grows when the edge is 5 cm.

12 For $s = t^n$ with $n \neq 0$, show that $\frac{as}{v^2}$ is the same constant at every instant.

13 A particle has $v = (1 + t)^2$ with $s(0) = 0$. Find $s(t)$, and the distance travelled in the first 2 seconds.

14 $C(x) = \frac{x^3}{3} - 10x^2 + 200x$. Find the output at which the marginal cost is least, and that least marginal cost.

15 A spherical balloon leaks at 2 cm³/s. Find the rate its surface area shrinks when the radius is 5 cm.

Revision · Control work 3 — the derivative in use

Tangents, turning points, optimisation, connected rates and approximation together — everything the derivative has been put to.

Lessons 35–36 · Control work 3 and work on the mistakes · 45 questions

KEY WORDS

- tangent
- stationary point
- optimisation
- connected rates
- approximation
- percentage error
- velocity
- acceleration
- justification

REMEMBER

1. Differentiate first, substitute second. Putting the number in early loses the function.
2. Connected rates: relate the quantities, differentiate with respect to t, substitute last.
3. Optimisation: objective, constraint, domain, $f' = 0$, then justify the nature.
4. $\Delta y \approx f'(x)\Delta x$ holds for a short step only — the error grows like $(\Delta x)^2$.
5. Check the units and the sense of every answer before writing it down.

Focus

EASY · 10 questions

The basic skill, one step at a time.

1. Differentiate $y = x^4$.

2. Differentiate $y = (2x + 1)^3$.

3. Find the tangent gradient to $y = x^2$ at $x = 5$.

4. $s = t^2 + t$. Find v at $t = 2$.

5. Estimate $(1.01)^2$.

6. Find the stationary point of $y = x^2 - 8x$.

7 Differentiate $y = \frac{2}{x^3}$.

8 Two numbers add to 20. Find their greatest product.

9 $s = 4t^3$. Find the acceleration.

10 Estimate $\sqrt{9.1}$.

Practice

MEDIUM 10 questions

The same skill with more to simplify first.

1 $y = x^3 - 12x$. Find the stationary points.

2 $y = x^3 - 12x$. Classify them.

3 Find the tangent to $y = x^3$ at $(2, 8)$.

4 A circle's radius grows at 3 cm/s. Find $\frac{dA}{dt}$ when $r = 4$ cm.

5 Estimate $(27.3)^{1/3}$.

6 A rectangle has perimeter 24 cm. Find its greatest area.

7 $y = \sqrt{x^2 + 9}$. Find $\frac{dy}{dx}$ at $x = 4$.

8 $s = t^3 - 3t$, with $t \geq 0$. Find when the particle is at rest.

9 A radius is measured with a 0.5 % error. Find the percentage error in the volume of the sphere.

10 $y = x + \frac{4}{x}$. Find the stationary points.

Challenge

HARD 10 questions

Letters in place of numbers, and more than one idea at once.

1 $y = x^2 - 2x + 3$. Find the tangent parallel to $y = 4x$.

2 A closed cylinder has volume 54π cm³. Find the radius giving least surface area.

3 Estimate $\sin 29°$ to four decimal places.

4 $s = t^4 - 4t^3$. Find every instant at which the acceleration is zero.

5 A ladder 10 m long has its foot 6 m from the wall, sliding out at 1 m/s. Find the rate the top falls.

6 $y = x^2(x - 4)$. Find and classify the stationary points.

7 Find the greatest value of $y = x(8 - x^2)$ on $[0,\ 2\sqrt{2}]$.

8 Estimate $(4.02)^{3/2}$.

9 The volume of a cube grows at 12 cm³/s. Find the rate the edge grows when the edge is 2 cm.

10 Show that $y = \dfrac{x - 1}{x + 2}$ is increasing on each of its branches.

Beyond

VERY HARD · 15 questions

Olympiad work. Expect to spend real time on these — some need an idea, not just a method.

1 $s = t^4 - 8t^2$. Find the total distance travelled in the first 3 seconds.

2 Find the greatest area of a rectangle with its base on the x-axis and its upper corners on $y = 9 - x^2$.

3 Show that the error in estimating $(1 + h)^n$ linearly is about $\frac{n(n-1)}{2}h^2$.

4 A trough 10 m long has a cross-section that is an isosceles triangle 1 m deep and 1 m across the top. Water flows in at 0.2 m³/min. Find the rate the depth rises when it is 0.5 m.

5 Prove that of all rectangles inscribed in a circle, the square has the greatest perimeter.

6 Find the shortest distance from $(0, 1)$ to the curve $y = x^2$.

7 A cone has fixed slant height L. Find its greatest volume.

8 Use the iteration $x_{n+1} = x_n - \dfrac{f(x_n)}{f'(x_n)}$ to solve $\cos x = x$ from $x_0 = 0.75$, to four decimal places.

9 Show that a cubic with a stationary point of inflection has exactly one real root.

10 A box with a square base and no lid has total surface area 108 cm². Find its greatest volume.

11 The period of a pendulum is $T = 2\pi\sqrt{\dfrac{L}{g}}$, with $g = 9.8$. Estimate the change in T when L goes from 1 m to 1.02 m.

12 $v = 3t^2 - 12t + 9$. Find the greatest speed on $0 \le t \le 4$.

13 Show that for a sphere $\frac{dV}{dr}$ equals the surface area, and say why that is to be expected.

14 Find a and b so that $y = ax^3 + bx$ has a stationary point at $(1, -2)$.

15 Two posts 6 m and 8 m high stand 10 m apart. A rope runs from the top of one to a point on the ground between them and up to the other. Find its least length.

The antiderivative and the indefinite integral

Differentiation run backwards — a whole family of curves, and the one condition that picks a single member out of it.

Lessons 37–40 · Algebra and Calculus 11, §2.1 · Pure Mathematics 1, 9.1 · 45 questions

KEY WORDS

- antiderivative
- indefinite integral
- constant of integration
- integrand
- family of curves
- initial condition
- particular solution
- integration
- inverse operation

REMEMBER

1. F is an antiderivative of f if $F' = f$. Checking one is always possible: differentiate it.
2. If F is an antiderivative then so is $F + C$, and every antiderivative has that form.
3. $\int f(x)dx = F(x) + C$. The $+ C$ is part of the answer, not decoration.
4. The family $F + C$ is a stack of parallel curves; an initial condition picks one out.
5. There is no product rule and no quotient rule for integration — simplify the integrand first.

Focus

EASY · 10 questions

The basic skill, one step at a time.

1. Find $\int x\,dx$.

2. Find $\int x^2\,dx$.

3. Find $\int 3\,dx$.

4. Find $\int x^5\,dx$.

5. Find $\int (2x + 1)\,dx$.

6. Find an antiderivative of $4x^3$.

7. Find $\int 0\ dx$.

8. Find $\int (x^2 - x)\ dx$.

9. $F'(x) = 2x$ and $F(0) = 5$. Find $F(x)$.

10. Differentiate $\frac{x^3}{3} + C$.

Practice

MEDIUM · 10 questions

The same skill with more to simplify first.

1. Find $\int (3x^2 - 4x + 1)\ dx$.

2. Find $\int \sqrt{x}\ dx$.

3. Find $\int x^{-2}\ dx$.

4. Find $\int (x + 1)^2\ dx$.

5. Find $\int \frac{1}{x^2}\ dx$.

6. $F'(x) = 3x^2$ and $F(1) = 4$. Find $F(x)$.

7 Find $\int(5 - 2x)\,dx$.

8 $v = 6t$ and $s(0) = 2$. Find $s(t)$.

9 Find $\int(x^3 + x)\,dx$.

10 A curve passes through $(1, 2)$ and has gradient $2x$. Find its equation.

Challenge

HARD 10 questions

Letters in place of numbers, and more than one idea at once.

1 Find $\int(x^2 + 1)^2\,dx$.

2 Find $\int \frac{x^2 - 1}{x^2}\,dx$.

3 Find $\int(\sqrt{x} + \frac{1}{\sqrt{x}})\,dx$.

4 Find $\int x(x + 2)\,dx$.

5 $F'(x) = \dfrac{1}{x^2}$ and $F(1) = 0$. Find $F(x)$.

6 $a = 6t$, with $v(0) = 0$ and $s(0) = 0$. Find $s(t)$.

7 Find $\int (2x + 3)^4 \, dx$.

8 Find $\int \dfrac{1}{\sqrt{x}} \, dx$.

9 A curve has gradient $3x^2 - 6x$ and passes through $(2, 0)$. Find its equation.

10 Find $\int (x + 1)(x - 1) \, dx$.

Beyond

VERY HARD · 15 questions

Olympiad work. Expect to spend real time on these — some need an idea, not just a method.

1 Prove that two antiderivatives of the same function differ by a constant.

2 Verify that $\int |x|\,dx = \frac{x|x|}{2} + C$.

3 Find f with $f'(x) = \frac{1}{x^2}$, $f(1) = 0$ and $f(-1) = 0$.

4 Show that $\int (1+x)^n\,dx = \frac{(1+x)^{n+1}}{n+1} + C$ for $n \neq -1$, and say what happens at $n = -1$.

5 A curve has $\frac{d^2y}{dx^2} = 6x$, with $\frac{dy}{dx} = 1$ and $y = 2$ at $x = 0$. Find y.

6 Show that $\int f'(x) \cdot f(x)\,dx = \frac{[f(x)]^2}{2} + C$.

7 Find every curve whose gradient at each point is twice the x-coordinate, and the one through $(2, 1)$.

8 A ball is thrown upwards at 20 m/s from ground level, with $a = -10$ m/s². Find its greatest height.

9 Explain why $\int x^n \, dx = \frac{x^{n+1}}{n + 1} + C$ fails for $n = -1$, and give the correct result.

10 Find $\int \frac{x^2 + 1}{x + 1} \, dx$.

11 Two curves have the same gradient function and meet at $(1, 3)$. Show that they are the same curve.

12 Find f with $f''(x) = 0$, $f(0) = 1$ and $f(1) = 4$.

13 A car starts from rest with $a = 12 - 6t$. Find when it is next at rest, and how far it has gone by then.

14 Describe the family $y = \frac{x^2}{2} + C$ geometrically, and say what every member shares.

15 F is an antiderivative of f and G of g. Show that $F + G$ is one of $f + g$, and that FG is generally not one of fg.

The table of integrals and the rules of integration

Every derivative read backwards, the linear substitution that extends them, and the rewriting that has to come first.

Lessons 41–46 · Algebra and Calculus 11, §2.2 · Pure Mathematics 1, 9.2–9.3 · 45 questions

KEY WORDS

- table of integrals
- power rule for integration
- linear substitution
- integrable form
- exceptional case
- term by term
- natural logarithm
- reciprocal function

REMEMBER

1. $\int x^n dx = \frac{x^{n+1}}{n+1} + C$ for $n \neq -1$; and $\int \frac{dx}{x} = \ln|x| + C$, the exceptional case.
2. $\int (ax+b)^n dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C$ — the $\frac{1}{a}$ is the mark most often lost.
3. $\int \sin x\, dx = -\cos x + C$ and $\int \cos x\, dx = \sin x + C$: the minus sign sits on the opposite one from differentiation.
4. Constants come out and sums split term by term. There is no product or quotient rule.
5. Rewrite first — roots and reciprocals as powers, products expanded, fractions split.

Focus

EASY · 10 questions

The basic skill, one step at a time.

1. Find $\int x^4\, dx$.

2. Find $\int \frac{dx}{x}$.

3. Find $\int e^x\, dx$.

4. Find $\int \sin x\, dx$.

5. Find $\int \cos x\, dx$.

6 Find $\int (2x + 1)^3 \, dx$.

7 Find $\int e^{2x} \, dx$.

8 Find $\int \frac{dx}{x + 1}$.

9 Find $\int \sec^2 x \, dx$.

10 Find $\int 5x^2 \, dx$.

Practice

MEDIUM 10 questions

The same skill with more to simplify first.

1 Find $\int (3x + 2)^5 \, dx$.

2 Find $\int \sin 3x \, dx$.

3 Find $\int \cos(2x + 1) \, dx$.

4 Find $\int e^{-x} \, dx$.

5 Find $\int \frac{dx}{2x + 3}$.

6 Find $\int \sqrt{x + 4}\, dx$.

7 Find $\int (x^2 + \frac{2}{x})\, dx$.

8 Find $\int 3^x\, dx$.

9 Find $\int \frac{dx}{(3x - 1)^2}$.

10 Find $\int (\sin x - \cos x)\, dx$.

Challenge

HARD 10 questions

Letters in place of numbers, and more than one idea at once.

1 Find $\int (x + 1)(x - 2)\, dx$.

2 Find $\int \frac{x^2 + 1}{x}\, dx$.

3 Find $\int \sin^2 x\, dx$.

4 Find $\int \cos^2 x \, dx$.

5 Find $\int \tan^2 x \, dx$.

6 Find $\int (e^x + 1)^2 \, dx$.

7 Find $\int \frac{dx}{\sqrt{2x + 1}}$.

8 Find $\int \frac{2x}{x^2 + 1} \, dx$.

9 Find $\int x(x^2 + 1)^3 \, dx$.

10 Find $\int (1 + \tan^2 x) \, dx$.

Beyond

VERY HARD · 15 questions

Olympiad work. Expect to spend real time on these — some need an idea, not just a method.

1. Prove the linear substitution rule $\int (ax+b)^n dx = \dfrac{(ax+b)^{n+1}}{a(n+1)} + C$ for $n \neq -1$.

2. Find $\int \sin x \cdot \cos x \, dx$ in three different ways, and reconcile the answers.

3. Find $\displaystyle\int \frac{dx}{x^2 - 1}$.

4. Find $\displaystyle\int \frac{x}{x+1}\, dx$.

5. Find $\displaystyle\int \frac{x^3}{x^2+1}\, dx$.

6 Find $\int \sin^3 x \, dx$.

7 Find $\int \frac{e^x}{1 + e^x} \, dx$.

8 Show that $\int \frac{f'(x)}{f(x)} \, dx = \ln|f(x)| + C$.

9 Find $\int \frac{\ln x}{x} \, dx$.

10 Find $\int \frac{dx}{x \ln x}$.

11 Find $\int |x - 1|\, dx$ on each side of $x = 1$, and show that the two pieces can be made to join.

12 Show that $\int \cos^2 x\, dx + \int \sin^2 x\, dx = x + C$, and check it against the Challenge answers.

13 Find $\int \dfrac{2x + 3}{x^2 + 3x + 1}\, dx$.

14 A curve has $\dfrac{dy}{dx} = \sin x$ and passes through $(\dfrac{\pi}{2}, 0)$. Find its equation.

15 Find $\int (e^x - e^{-x})\, dx$, and say what the answer is a multiple of.

Revision · Control work 4 — the indefinite integral

The antiderivative, the table and the rules together — and the quarter behind them.

Lessons 47–48 · Control work 4 and the quarter review · 45 questions

KEY WORDS

- antiderivative
- indefinite integral
- constant of integration
- linear substitution
- initial condition
- term by term
- rewrite first
- check by differentiating

REMEMBER

1. Rewrite before integrating: expand, split, turn roots and reciprocals into powers.
2. Linear inner function: divide by a. Anything harder is Quarter III.
3. Every indefinite integral ends in $+ C$; every initial condition pins it down.
4. Reversing a second derivative needs two conditions, not one.
5. Differentiate your answer. It takes seconds and it settles the question.

Focus

EASY · 10 questions

The basic skill, one step at a time.

1. Find $\int x^3 \, dx$.

2. Find $\int dx$.

3. Find $\int e^{4x} \, dx$.

4. Find $\int \cos 4x \, dx$.

5. Find $\int (x + 2) \, dx$.

6. Find $\int \dfrac{dx}{x + 3}$.

7 Find $\int (3x - 1)^2 \, dx$.

8 Find $\int \sin(x + 1) \, dx$.

9 $F'(x) = 4x$ and $F(0) = 1$. Find $F(x)$.

10 Find $\int 2^x \, dx$.

Practice

MEDIUM 10 questions

The same skill with more to simplify first.

1 Find $\int (x^2 - 3x + 2) \, dx$.

2 Find $\int (4x + 1)^3 \, dx$.

3 Find $\int x\sqrt{x} \, dx$.

4 Find $\int e^{3x} \, dx$.

5 Find $\int \frac{dx}{x - 2}$.

6 Find $\int \sin 2x \, dx$.

7 $v = 4t + 1$ and $s(0) = 0$. Find $s(t)$.

8 Find $\int \frac{x - 1}{x}\,dx$.

9 Find $\int \sec^2 3x\,dx$.

10 A curve has gradient $\frac{1}{x^2}$ and passes through $(1, 2)$. Find its equation.

Challenge

HARD 10 questions

Letters in place of numbers, and more than one idea at once.

1 Find $\int \frac{2x + 1}{x^2 + x + 1}\,dx$.

2 Find $\int (x + 1)^3\,dx$.

3 Find $\int (1 - x)^5\,dx$.

4 Find $\int (\sin x + \sec^2 x)\,dx$.

5 Find $\int x^2(x^3 + 1)^2\,dx$.

6 Find $\int \dfrac{dx}{1 - 2x}$.

7 Find $\int (e^x + e^{2x})\,dx$.

8 $a = 2$, with $v(0) = 3$ and $s(0) = 1$. Find $s(t)$.

9 Find $\int (\sqrt{x} - 1)^2\,dx$.

10 Find $\int \cos^2 \dfrac{x}{2}\,dx$.

Beyond

VERY HARD · 15 questions

Olympiad work. Expect to spend real time on these — some need an idea, not just a method.

1 Find $\int x\sqrt{x + 1}\,dx$ by writing $u = x + 1$.

2 Find $\int \frac{dx}{e^x + 1}$.

3 Find $\int \sin^4 x \, dx$.

4 Verify by differentiation that $\int \sec x \, dx = \ln|\sec x + \tan x| + C$.

5 A ball is thrown upwards at 29.4 m/s from a height of 10 m, with $a = -9.8$ m/s². Find its greatest height.

6 Find $\int \frac{x^2 + 1}{(x + 1)^2} \, dx$.

7 Show that if f is odd then the antiderivative F with $F(0) = 0$ is even.

8 Find $\int \frac{dx}{x^2 + 4}$.

9 Verify by differentiation that $\int x\, e^x\, dx = e^x(x - 1) + C$.

10 A curve has $\frac{d^2y}{dx^2} = \sin x$, with $\frac{dy}{dx} = 0$ and $y = 0$ at $x = 0$. Find y.

11 Find $\int \frac{1 + x}{1 + x^2}\, dx$.

12 Show that $\int [f(x)]^n f'(x)\,dx = \dfrac{[f(x)]^{n+1}}{n+1} + C$ for $n \neq -1$, and use it on $\int x^2(x^3+5)^4\,dx$.

13 Water flows into an empty tank at $\dfrac{dV}{dt} = 6t - t^2$ litres per minute. Find the volume when the flow stops.

14 Find $\int \dfrac{dx}{\sqrt{4-x^2}}$.

15 Explain why $\int e^{x^2} dx$ has no answer in elementary functions, and say what is done instead.

Answers

Topic 1 · Increments, and the problem of the tangent

Focus · easy

1. 9
2. 15
3. 5
4. 1
5. $6h + h^2$
6. $6 + h$
7. 0
8. 4
9. 4
10. 60 km/h

Practice · medium

1. $h^2 + 4h$
2. $h + 4$
3. $-\dfrac{h}{2(2+h)}$
4. $-\dfrac{1}{2(2+h)}$
5. $3h + 3h^2 + h^3$
6. $3 + 3h + h^2$
7. $-\dfrac{1}{2}$
8. 4
9. 5
10. $\sqrt{4+h} - 2$

Challenge · hard

1. $2a + h$
2. $-\dfrac{2+h}{(1+h)^2}$
3. $\dfrac{1}{\sqrt{a+h}+\sqrt{a}}$
4. $2ax + b + ah$
5. $h = 1$
6. $-\dfrac{1}{ab}$
7. $10 + 5h$
8. 10 — the speed at the instant $t = 1$
9. $3a^2 + 3ah + h^2$
10. $\dfrac{h}{h+1}$

Beyond · very hard

1 na^{n-1}

By the binomial theorem $(a+h)^n = a^n + na^{n-1}h + \dfrac{n(n-1)}{2}a^{n-2}h^2 + \ldots + h^n$. Subtracting a^n and dividing by h leaves $na^{n-1} + \dfrac{n(n-1)}{2}a^{n-2}h + \ldots + h^{n-1}$. Every term after the first carries a factor h, so as $h \to 0$ they all vanish and the quotient tends to na^{n-1}.

2 $x = 2$

The secant gradient is $\dfrac{9-1}{3-1} = 4$. The tangent gradient at $x = c$ is $2c$, so $2c = 4$ and $c = 2$, which lies in $(1, 3)$. This is the mean value theorem in its simplest case.

3 True for every quadratic

Let $f(x) = px^2 + qx + r$. Then $f(b) - f(a) = p(b^2 - a^2) + q(b - a) = (b - a)[p(a + b) + q]$, so the secant gradient is $p(a + b) + q$. The tangent gradient at $x = c$ is $2pc + q$, and $2pc + q = p(a + b) + q$ gives $c = \dfrac{a+b}{2}$.

4 $c = \sqrt{ab}$

The secant gradient is $-\frac{1}{ab}$ from Challenge 6. The tangent gradient at $x = c$ is $-\frac{1}{c^2}$, so $c^2 = ab$ and $c = \sqrt{ab}$, the geometric mean. It lies between a and b because $a < \sqrt{ab} < b$ whenever $0 < a < b$.

5 0; no tangent at the corner

$f(1) = f(-1) = 1$, so $\Delta y = 0$ and the average rate of change is 0. At $x = 0$ the quotient $\frac{|h| - 0}{h}$ equals 1 for $h > 0$ and -1 for $h < 0$. The two one-sided limits disagree, so the secant has no single limiting position and no tangent exists.

6 a^2

$f(a) - f(-a) = a^3 - (-a)^3 = 2a^3$, and the run is $a - (-a) = 2a$. The gradient is $\frac{2a^3}{2a} = a^2$, for every $a \neq 0$.

7 $2p$

$\frac{q^2 - p^2}{q - p} = \frac{(q - p)(q + p)}{q - p} = p + q$. Letting $q \to p$ slides the second point onto the first, and the secant gradient $p + q$ tends to $2p$. So the tangent gradient at $x = p$ is $2p$.

8 $a = -1$

By Challenge 1 with $h = 2$ the secant gradient is $2a + 2$. Setting $2a + 2 = 0$ gives $a = -1$, so the interval is $[-1, 1]$ — symmetric about the vertex, which is the only way a parabola can return to the same height.

9 $3h + h^2$; limit 0 — the point is at rest at $t = 1$

$s(1) = -2$ and $s(1 + h) = (1 + h)^3 - 3(1 + h) = -2 + 3h^2 + h^3$. So $\Delta s = 3h^2 + h^3$ and $\frac{\Delta s}{h} = 3h + h^2$, which tends to 0. The point is momentarily at rest at $t = 1$, where it turns round.

10 $\frac{1}{\sqrt{a} + \sqrt{b}}$

The gradient is $\frac{\sqrt{b} - \sqrt{a}}{b - a}$. Multiply numerator and denominator by the conjugate $\sqrt{b} + \sqrt{a}$: the numerator becomes $b - a$, which cancels with the denominator and leaves $\frac{1}{\sqrt{a} + \sqrt{b}}$.

11 Both equal $2t + 2$

The secant gradient is $2t + 2$ by Challenge 1 with $h = 2$. The tangent gradient at $x = t + 1$ is $2(t + 1) = 2t + 2$. They agree — the midpoint property of question 3 again.

12 $b = -3$

$f(0) = 0$ and $f(4) = 16 + 4b$, so the secant gradient is $\frac{16 + 4b}{4} = 4 + b$. Setting $4 + b = 1$ gives $b = -3$.

13 Always negative

The gradient is $-\frac{1}{ab}$. With $a > 0$ and $b > 0$ the product ab is positive, so $-\frac{1}{ab} < 0$. Geometrically the branch in the first quadrant falls everywhere, so every chord of it slopes downwards.

14 $f(x) = x^2 + x$

The average rate of change is $\frac{f(x) - f(0)}{x - 0} = \frac{f(x)}{x}$. Setting this equal to $x + 1$ gives $f(x) = x(x + 1) = x^2 + x$. Checking: $f(0) = 0$, as required.

15 4 — the same limit

$f(2 + h) = 4 + 4h + h^2$ and $f(2 - h) = 4 - 4h + h^2$, so the numerator is $8h$ and the quotient is exactly 4 for every $h \neq 0$. The limit is 4, the same as the ordinary quotient gives. This symmetric quotient straddles the point instead of leaning to one side, which is why it is the one a calculator uses.

Topic 2 · The limit of a function

Focus · easy

1. 7
2. -4
3. 6
4. 2
5. $-\dfrac{1}{4}$
6. 2
7. 5
8. 0
9. 12
10. 1

Practice · medium

1. 1
2. $\dfrac{1}{6}$
3. 3
4. 0
5. The limit does not exist — the quotient grows without bound
6. 1
7. $\dfrac{3}{2}$
8. -1
9. 4
10. 3

Challenge · hard

1. $\dfrac{5}{2}$
2. $\dfrac{1}{2}$
3. 0
4. $\dfrac{1}{2}$
5. 1
6. $\dfrac{9}{4}$
7. e
8. 1
9. $a = 4$
10. -1 and 1; the limit does not exist

Beyond · very hard

1 $\dfrac{1}{3}$

Put $u = (1 + x)^{1/3}$, so $u^3 = 1 + x$ and $x = u^3 - 1$. As $x \to 0$, $u \to 1$. The quotient becomes $\dfrac{u - 1}{u^3 - 1} = \dfrac{u - 1}{(u - 1)(u^2 + u + 1)} = \dfrac{1}{u^2 + u + 1}$, which tends to $\dfrac{1}{3}$.

2 n

$x^n - 1 = (x - 1)(x^{n-1} + x^{n-2} + \ldots + x + 1)$. Cancelling $x - 1$ leaves a sum of n terms, each tending to 1 as $x \to 1$, so the limit is n. Notice this is the derivative of x^n at $x = 1$.

3 $\dfrac{a}{b}$

Write the quotient as $\dfrac{\sin ax}{ax} \cdot \dfrac{bx}{\sin bx} \cdot \dfrac{a}{b}$. The first factor tends to 1 and the second to 1, leaving $\dfrac{a}{b}$.

4 $\frac{1}{2}$

Multiply by the conjugate: $\frac{x}{\sqrt{x^2+x}+x}$. Divide top and bottom by x (positive, so $x=\sqrt{x^2}$): $\frac{1}{\sqrt{1+1/x}+1}$, which tends to $\frac{1}{2}$.

5 $\frac{1}{2}$

$\tan x-\sin x=\sin x(\frac{1}{\cos x}-1)=\frac{\sin x(1-\cos x)}{\cos x}$. Dividing by x^3 gives $\frac{\sin x}{x}\cdot\frac{1-\cos x}{x^2}\cdot\frac{1}{\cos x}$, which tends to $1\cdot\frac{1}{2}\cdot 1=\frac{1}{2}$.

6 2

Use $1-\cos 2x=2\sin^2 x$. The quotient becomes $\frac{2\sin^2 x}{x\cdot\sin x}=\frac{2\sin x}{x}$, which tends to 2.

7 $\frac{1}{4}$

Multiply by the conjugate $\sqrt{x+2}+2$. The numerator becomes $(x+2)-4=x-2$, which cancels the denominator and leaves $\frac{1}{\sqrt{x+2}+2}$, tending to $\frac{1}{4}$.

8 e

Put $t=\frac{1}{x}$. As $x\to 0^+$, $t\to\infty$, and $(1+x)^{1/x}$ becomes $(1+\frac{1}{t})^t$, whose limit is e. The same substitution handles $x\to 0^-$ with $t\to-\infty$, and that limit is e too. They are the same limit written in two variables.

9 0

For every $x\neq 0$, $-1\le\sin\frac{1}{x}\le 1$, so $-|x|\le x\cdot\sin\frac{1}{x}\le|x|$. Both bounds tend to 0 as $x\to 0$, so the middle expression is squeezed to 0. Note that $\sin\frac{1}{x}$ itself has no limit at 0 — the factor x is what kills the oscillation.

10 1

Put $t=\frac{1}{x}$, so $t\to 0$ as $x\to\infty$. The expression becomes $\frac{\sin t}{t}$, whose limit is 1. Contrast with the previous question: the same two factors, opposite limits.

11 $a=1,\ b=-1$

Divide: $\frac{x^2+1}{x+1}=x-1+\frac{2}{x+1}$. Subtracting $ax+b$ leaves $(1-a)x-(1+b)+\frac{2}{x+1}$. The last term vanishes on its own, so the limit is 0 exactly when $1-a=0$ and $1+b=0$. The line $y=x-1$ is the oblique asymptote.

12 -1

Since $1-x^3=(1-x)(1+x+x^2)$, the common denominator is $(1-x)(1+x+x^2)$ and the numerator becomes $(1+x+x^2)-3=x^2+x-2=(x+2)(x-1)$. So the expression is $\frac{(x+2)(x-1)}{-(x-1)(1+x+x^2)}=-\frac{x+2}{1+x+x^2}$, which tends to $-\frac{3}{3}=-1$.

13 $\dfrac{3}{2}$

Divide numerator and denominator by x. The numerator over x tends to 1 by Challenge 5. For the denominator, question 1 gives $\dfrac{(1+x)^{1/3}-1}{x} \to \dfrac{1}{3}$, and the same with $-x$ gives $\dfrac{1-(1-x)^{1/3}}{x} \to \dfrac{1}{3}$; adding, the denominator over x tends to $\dfrac{2}{3}$. The limit is $1 \div \dfrac{2}{3} = \dfrac{3}{2}$.

14 $\dfrac{\pi}{180}$

An angle of x degrees is $\dfrac{\pi x}{180}$ radians, so $\sin x° = \sin\dfrac{\pi x}{180}$. Then $\dfrac{\sin x°}{x} = \dfrac{\pi}{180} \cdot \dfrac{\sin(\pi x/180)}{\pi x/180}$, and the second factor tends to 1. The limit is $\dfrac{\pi}{180} \approx 0.01745$. This is exactly why calculus insists on radians: only there does the constant come out as 1.

15 Both limits are 2

Near $x = 1$ we have $x > 0$, so $|x| = x$ and $f(x) = \dfrac{(x-1)(x+1)}{x-1} = x + 1 \to 2$. Near $x = -1$ we have $x < 0$, so $|x| = -x$ and $f(x) = \dfrac{(x-1)(x+1)}{-(x+1)} = 1 - x \to 2$. Both points are removable discontinuities, and both would be repaired by the value 2.

Topic 3 · The derivative of a function

Focus · easy

1 0

2 4

3 3

4 $2x$

5 10

6 $3x^2$

7 $2x$

8 6

9 8

10 $\dfrac{dy}{dx} = 2x$

Practice · medium

1 $4x - 3$

2 $-\dfrac{1}{x^2}$

3 $-\dfrac{1}{4}$

4 $\dfrac{1}{2\sqrt{x}}$

5 $\dfrac{1}{6}$

6 $x = 2$

7 -1

8 8

9 11

10 $-\dfrac{2}{x^3}$

Challenge · hard

1 $-\dfrac{1}{(x+1)^2}$

6 0

7 $-\dfrac{1}{2x\sqrt{x}}$

2 $\dfrac{1}{2\sqrt{x+3}}$

3 $2ax + b$

4 $4x^3$

5 The one-sided derivatives are 1 and -1

8 $t = 0$ and $t = 4$

9 $x = 3.5$

10 $x = 2$ and $x = -2$

Beyond · very hard

1 nx^{n-1}

Expand $(x+h)^n$ by the binomial theorem, subtract x^n and divide by h: the result is $nx^{n-1} + \dfrac{n(n-1)}{2}x^{n-2}h + \dots + h^{n-1}$. Every term beyond the first carries a factor h and vanishes as $h \to 0$, leaving nx^{n-1}.

2 $\dfrac{1}{3x^{2/3}}$

Write $a = (x+h)^{1/3}$ and $b = x^{1/3}$, so $a^3 - b^3 = h$. Using $a^3 - b^3 = (a-b)(a^2+ab+b^2)$, the quotient $\dfrac{a-b}{h}$ equals $\dfrac{1}{a^2+ab+b^2}$. As $h \to 0$ we get $a \to b$, so the denominator tends to $3b^2 = 3x^{2/3}$.

3 $\cos x$

$\sin(x+h) - \sin x = \sin x \cdot \cos h + \cos x \cdot \sin h - \sin x$. Dividing by h gives $\cos x \cdot \dfrac{\sin h}{h} - \sin x \cdot \dfrac{1-\cos h}{h}$. The first factor tends to 1 and the second to 0, so the limit is $\cos x$.

4 $-\sin x$

$\cos(x+h) - \cos x = \cos x \cdot \cos h - \sin x \cdot \sin h - \cos x$. Dividing by h gives $-\sin x \cdot \dfrac{\sin h}{h} - \cos x \cdot \dfrac{1-\cos h}{h}$, which tends to $-\sin x$. The minus sign is the one every class loses.

5 Differentiability implies continuity

Write $f(x) - f(a) = \dfrac{f(x)-f(a)}{x-a} \cdot (x-a)$ for $x \neq a$. As $x \to a$ the first factor tends to $f'(a)$, a finite number, and the second tends to 0. The product tends to 0, so $f(x) \to f(a)$, which is continuity.

6 Yes, and $f'(1) = 2$

Both pieces give $f(1) = 1$, so f is continuous. The left derivative is $2x$ at $x = 1$, that is 2; the right derivative is 2. They agree, so $f'(1) = 2$. The line is exactly the tangent to the parabola at that point, which is why the join is smooth.

7 No — continuous but not differentiable

Both pieces give $f(1) = 1$, so f is continuous at 1. But the left derivative is 2 and the right derivative is 3. They disagree, so there is a corner and no derivative. Continuity is not enough.

8 0

The quotient is $\dfrac{h^2 \sin(1/h) - 0}{h} = h \cdot \sin\dfrac{1}{h}$, and that is squeezed to 0 between $-|h|$ and $|h|$. So $f'(0) = 0$, even though f' itself oscillates wildly near 0 and has no limit there.

9 $a = 4$, $b = -4$

Matching the derivatives gives $a = 2(2) = 4$. Matching the values gives $4 = 2a + b = 8 + b$, so $b = -4$. The line is $y = 4x - 4$, the tangent to $y = x^2$ at $(2, 4)$ — the only line that can join on smoothly.

10 $a = 1$ and $a = -1$

The tangent at $x = a$ is $y - a^2 = 2a(x-a)$, that is $y = 2ax - a^2$. Putting $x = 0$ gives $y = -a^2$, and this must equal -1, so $a^2 = 1$ and $a = \pm 1$. Two tangents pass through that point.

11 Only at $(0, 0)$

A horizontal tangent needs $f'(x) = 3x^2 = 0$, so $x = 0$ and the point is $(0, 0)$. This is a stationary point that is not a maximum or a minimum — the curve flattens and then carries on rising, which is a stationary point of inflection.

12 $f'(x) = 2|x|$

For $x > 0$, $f(x) = x^2$ and $f'(x) = 2x = 2|x|$. For $x < 0$, $f(x) = -x^2$ and $f'(x) = -2x = 2|x|$. At 0 the quotient is $\frac{h|h|}{h} = |h| \to 0 = 2|0|$. So the formula holds everywhere, and f' is continuous — unlike $|x|$ itself, this function is smooth at the origin.

13 $v = 3t^2 - 6t$; at rest at $t = 0$ and $t = 2$

Differentiating gives $v(t) = 3t^2 - 6t = 3t(t - 2)$, which is zero at $t = 0$ and $t = 2$. Between those instants $v < 0$, so the particle moves backwards before turning round again at $t = 2$.

14 No derivative — a cusp

$f(x) = x^{2/3}$, so the quotient at 0 is $\frac{h^{2/3}}{h} = h^{-1/3}$. As $h \to 0^+$ this tends to $+\infty$ and as $h \to 0^-$ to $-\infty$. The two sides run off in opposite directions: the curve has a vertical cusp at the origin, not a tangent of finite gradient.

15 It touches only at $x = a$

The tangent is $y = 2ax - a^2$. Setting $x^2 = 2ax - a^2$ gives $x^2 - 2ax + a^2 = (x - a)^2 = 0$, whose only root is $x = a$, repeated. The repeated root is the algebraic signature of touching rather than crossing — and it is special to the parabola, since a tangent to a cubic generally does meet it again.

Topic 4 · The rules of differentiation

Focus · easy

1 $5x^4$

2 $12x^3$

3 $2x + 1$

4 0

5 $-3x^{-4}$

6 $\frac{1}{2\sqrt{x}}$

7 $-\frac{1}{x^2}$

8 $12x^2 - 4x$

9 $5x^4$

10 3

Practice · medium

1 $2x + 3$

2 $3x^2 - 6x$

3 $1 - \frac{1}{x^2}$

4 $\frac{3\sqrt{x}}{2}$

5 $-\frac{6}{x^3}$

6 $4x + 7$

7 $1 + \frac{2}{x^3}$

8 $\frac{5}{2\sqrt{x}} + \frac{2}{x^2}$

9 $2x + 2$

10 $12x^2$

Challenge · hard

1 $5x^4 + 3x^2 - 4$

2 $\frac{1}{(x + 1)^2}$

6 1

7 $x^{-1/2} - \frac{1}{2}x^{-3/2}$

3 $\dfrac{2}{(x+1)^2}$

4 $\dfrac{x^2-2x-1}{(x-1)^2}$

5 $\dfrac{7}{2}x^{5/2}$

8 $\dfrac{2x}{(x^2+1)^2}$

9 $\dfrac{2}{x^3}$

10 -1

Beyond · very hard

1 Proved

The increment of uv is $u(x+h)v(x+h) - u(x)v(x)$. Add and subtract $u(x+h)v(x)$: it becomes $u(x+h)[v(x+h) - v(x)] + v(x)[u(x+h) - u(x)]$. Dividing by h and letting $h \to 0$, the first bracket gives $u(x)v'(x)$ — using that u is continuous — and the second gives $v(x)u'(x)$.

2 Proved

Write $u = (\dfrac{u}{v}) \cdot v$ and let $q = \dfrac{u}{v}$. The product rule gives $u' = q'v + qv'$, so $q'v = u' - \dfrac{u}{v}v' = \dfrac{u'v - uv'}{v}$.

Dividing by v gives $q' = \dfrac{u'v - uv'}{v^2}$.

3 Proved

Apply the product rule to $(uv) \cdot w$: the derivative is $(uv)'w + (uv)w'$. Expanding $(uv)' = u'v + uv'$ gives $u'vw + uv'w + uvw'$. The pattern continues: differentiate each factor in turn and leave the others alone.

4 $-\dfrac{v'}{v^2}$

Take $u = 1$ in the quotient rule. Then $u' = 0$ and $(\dfrac{1}{v})' = \dfrac{0 \cdot v - 1 \cdot v'}{v^2} = -\dfrac{v'}{v^2}$. This is the rule worth memorising separately, since it turns up far more often than the full quotient rule.

5 $3x^2 - 12x + 11$

Expanding first is quickest: $(x-1)(x-2) = x^2 - 3x + 2$, and multiplying by $x - 3$ gives $x^3 - 6x^2 + 11x - 6$. Differentiating term by term gives $3x^2 - 12x + 11$. The triple product rule gives the same answer with more work.

6 $x = 1$ and $x = -1$

$\dfrac{dy}{dx} = 3x^2 - 3 = 3(x^2 - 1) = 3(x-1)(x+1)$, which vanishes at $x = \pm 1$. These are the local maximum at $x = -1$ and the local minimum at $x = 1$.

7 8

By the product rule $f'(x) = 2x \cdot g(x) + x^2 \cdot g'(x)$. At $x = 2$ this is $4 \cdot 3 + 4 \cdot (-1) = 12 - 4 = 8$. The formula for g is never needed — only its value and its derivative at the one point.

8 $a = 3$

The quotient rule gives $f'(x) = \dfrac{2x(x+1) - (x^2 + a)}{(x+1)^2}$. At $x = 1$ the numerator is $4 - (1 + a) = 3 - a$, and a stationary point needs it to be zero, so $a = 3$.

9 Only at $x = \pm 1$

$\dfrac{dy}{dx} = 1 - \dfrac{1}{x^2}$, which is zero when $x^2 = 1$, that is $x = \pm 1$. Both lie in the domain $x \neq 0$. The point $x = 1$ gives the minimum value 2 on the right branch and $x = -1$ the maximum -2 on the left.

10 $\dfrac{4}{(x-1)^3}$

The quotient rule gives $y' = \dfrac{(x-1)-(x+1)}{(x-1)^2} = \dfrac{-2}{(x-1)^2} = -2(x-1)^{-2}$. Differentiating again, $y'' = 4(x-1)^{-3} = \dfrac{4}{(x-1)^3}$.

11 f' is odd

An even function satisfies $f(-x) = f(x)$. Differentiating both sides with respect to x, the left gives $-f'(-x)$ by the chain rule and the right gives $f'(x)$. So $f'(-x) = -f'(x)$, which is exactly the definition of an odd function. Picture it: an even graph is symmetric about the y-axis, so matching points have opposite slopes.

12 f' is even

An odd function satisfies $f(-x) = -f(x)$. Differentiating, the left gives $-f'(-x)$ and the right $-f'(x)$, so $f'(-x) = f'(x)$ and f' is even. A consequence worth noticing: the derivative of an odd function at 0, if it exists, need not be zero — but the function itself must pass through the origin.

13 $f^{(n)}(x) = n!$

Each differentiation drops the index by one and multiplies by the old index: after one step nx^{n-1}, after two $n(n-1)x^{n-2}$, and after k steps $n(n-1)\ldots(n-k+1)x^{n-k}$. Taking $k = n$ leaves $n(n-1)\ldots 1 \cdot x^0 = n!$. One further differentiation gives 0.

14 Only $f(x) = 0$

Suppose f has degree $n \geq 1$. Then f' has degree $n-1$, and a polynomial cannot equal one of a different degree, so no such f exists. If f is a constant c then $f' = 0$, and $0 = c$ forces $c = 0$. So the zero polynomial is the only solution — the equation $y' = y$ needs e^x, which is not a polynomial.

15 $a = -12$, $b = 1$

$\dfrac{dy}{dx} = 3x^2 + a$, and a stationary point at $x = 2$ needs $12 + a = 0$, so $a = -12$. Passing through $(0, 1)$ gives $b = 1$. The curve is $y = x^3 - 12x + 1$, with its other stationary point at $x = -2$.

Topic 5 · The derivative of a composite function

Focus · easy

1 $5(x+1)^4$

2 $8(2x+3)^3$

3 $6(3x-1)$

4 $6x(x^2+1)^2$

5 $\dfrac{1}{2\sqrt{x+4}}$

6 $375x^2$

7 $-6(1-x)^5$

8 $6x^2(x^3+2)$

9 $-\dfrac{1}{(x+1)^2}$

10 $40(4x-7)^9$

Practice · medium

1 $\dfrac{x}{\sqrt{x^2+1}}$

2 $4(2x-3)(x^2-3x)^3$

3 $-\dfrac{6}{(2x+1)^4}$

4 $\dfrac{3(1+\sqrt{x})^2}{2\sqrt{x}}$

6 $2(x+\dfrac{1}{x})(1-\dfrac{1}{x^2})$

7 $-10x(2-x^2)^4$

8 $-\dfrac{1}{2(x+1)^{3/2}}$

9 0

10 $-\dfrac{4(x+1)}{(x-1)^3}$

5 $\dfrac{3}{2\sqrt{3x-2}}$

Challenge · hard

1 $2x(2x+1)^2(5x+1)$

2 $\dfrac{3x+2}{2\sqrt{x+1}}$

3 $\dfrac{2(x+1)^2(x-2)}{(x-1)^2}$

4 $\dfrac{2x+1}{2\sqrt{x^2+x+1}}$

5 $\dfrac{1+\sqrt{1+x}}{\sqrt{1+x}}$

6 $\dfrac{15(2x-1)^2}{(x+2)^4}$

7 512

8 0

9 $(3x+1)^{-2/3}$

10 20π cm²/s

Beyond · very hard

1 The chain rule

For a small Δx write $\dfrac{\Delta y}{\Delta x} = \dfrac{\Delta y}{\Delta u}\cdot\dfrac{\Delta u}{\Delta x}$, valid whenever $\Delta u \neq 0$. As $\Delta x \to 0$ the continuity of u forces $\Delta u \to 0$, so the first factor tends to $\dfrac{dy}{du}$ and the second to $\dfrac{du}{dx}$. A full proof has to handle the case $\Delta u = 0$ separately, which is why the textbook states the rule rather than proving it here.

2 $12x(x^2+1)^2((x^2+1)^3+1)$

Work from the outside in. The outer square gives $2((x^2+1)^3+1)$; the cube gives $3(x^2+1)^2$; the innermost gives $2x$. Multiplying the three factors gives $12x(x^2+1)^2((x^2+1)^3+1)$.

3 $\dfrac{1}{2\sqrt{x+\sqrt{x}}}\left(1+\dfrac{1}{2\sqrt{x}}\right)$

The outer function is a square root, so its derivative is $\dfrac{1}{2\sqrt{x+\sqrt{x}}}$. The inner function $x+\sqrt{x}$ has derivative $1+\dfrac{1}{2\sqrt{x}}$. Multiply the two. A two-link chain where the inner link is itself a sum, not a composite.

4 6

By the chain rule $\dfrac{dy}{dx} = f'(x^2)\cdot 2x$. At $x=1$ this is $f'(1)\cdot 2 = 3\cdot 2 = 6$. The formula for f is never needed.

5 -6

$\dfrac{dy}{dx} = 2f(x)\cdot f'(x)$, so at $x=2$ it is $2\cdot 3\cdot(-1) = -6$. Compare with the previous question: there the function was inside, here it is outside.

6 $\dfrac{1}{\pi}$ cm/s

$V = \dfrac{4}{3}\pi r^3$, so $\dfrac{dV}{dr} = 4\pi r^2$. The chain rule gives $\dfrac{dV}{dt} = 4\pi r^2\cdot\dfrac{dr}{dt}$. At $r=5$, $100 = 100\pi\cdot\dfrac{dr}{dt}$, so $\dfrac{dr}{dt} = \dfrac{1}{\pi} \approx 0.318$ cm/s.

7 $n!a^n$

Each differentiation drops the index by one and brings down the current index together with a factor a from the chain rule. After k steps the result is $n(n-1)\ldots(n-k+1)a^k(ax+b)^{n-k}$. Taking $k=n$ leaves $n!a^n$, with no x remaining.

8 $\dfrac{1}{(1+x^2)^{3/2}}$

The quotient rule gives $\dfrac{\sqrt{1+x^2} - x\cdot\dfrac{x}{\sqrt{1+x^2}}}{1+x^2}$. Multiplying numerator and denominator by $\sqrt{1+x^2}$ turns the numerator into $(1+x^2) - x^2 = 1$, leaving $\dfrac{1}{(1+x^2)^{3/2}}$. The derivative is positive everywhere, so the function is always increasing.

9 0.375 m/s downwards

With x along the ground and y up the wall, $x^2 + y^2 = 25$. Differentiating with respect to t: $2x\dfrac{dx}{dt} + 2y\dfrac{dy}{dt} = 0$. When $x = 3$, $y = 4$, so $\dfrac{dy}{dt} = -\dfrac{x}{y}\dfrac{dx}{dt} = -\dfrac{3\times 0.5}{4} = -0.375$. The minus sign says the top is descending.

10 n

$\dfrac{dy}{dx} = n(1+x)^{n-1}$, which is n at $x = 0$. The binomial expansion begins $1 + nx + \dots$, and the coefficient of x in an expansion about 0 is exactly the derivative there. The two statements are the same fact.

11 8

$g'(x) = 2f'(2x+1)$. At $x = 2$ the inner value is $2(2) + 1 = 5$, so $g'(2) = 2f'(5) = 2 \times 4 = 8$.

12 $\dfrac{1}{4\sqrt{1+x}\sqrt{1+\sqrt{1+x}}}$

Outer root: $\dfrac{1}{2\sqrt{1+\sqrt{1+x}}}$. Inner root: $\dfrac{1}{2\sqrt{1+x}}$. Multiplying gives $\dfrac{1}{4\sqrt{1+x}\cdot\sqrt{1+\sqrt{1+x}}}$. A three-link chain, with the innermost link $x + 1$ contributing a factor 1.

13 Verified

$\dfrac{dy}{dx} = \dfrac{x}{\sqrt{x^2+1}}$, so $y\dfrac{dy}{dx} = \sqrt{x^2+1}\cdot\dfrac{x}{\sqrt{x^2+1}} = x$. The same identity follows at once from $y^2 = x^2 + 1$ by differentiating both sides: $2y\,y' = 2x$.

14 360 cm²/s

Surface area $S = 6a^2$, so $\dfrac{dS}{dt} = 12a\dfrac{da}{dt}$. At $a = 10$ with $\dfrac{da}{dt} = 3$ this is $12 \times 10 \times 3 = 360$ cm²/s. Note the rate depends on the current edge, so it is not constant even though the edge grows steadily.

15 y is decreasing everywhere

By the chain rule $\dfrac{dy}{dx} = f'(g(x))\cdot g'(x)$. The first factor is positive, being a value of f', and the second is negative. A positive times a negative is negative, so $\dfrac{dy}{dx} < 0$ everywhere and y decreases. An increasing function of a decreasing one reverses the direction.

Topic 6 · Revision · Control work 1 — the derivative

Focus · easy

1 4

2 4

3 $7x^6$

4 $15x^2 - 2$

6 $\dfrac{1}{4}$

7 $4(x+2)^3$

8 $6x$

9 $2x + 1$

5 $-\dfrac{3}{x^4}$

10 $6x$

Practice · medium

1 $\dfrac{1}{4}$

2 $\dfrac{2}{3}$

3 $4x + 5$

4 $\dfrac{1 - x^2}{(x^2 + 1)^2}$

5 $\dfrac{1}{\sqrt{2x + 1}}$

6 $6x(x^2 - 1)^2$

7 $-\dfrac{2}{x^2}$

8 $2x - \dfrac{1}{x^2}$

9 11

10 $x = 3$

Challenge · hard

1 $\dfrac{4}{3}$

2 $\dfrac{x(5x + 4)}{2\sqrt{x + 1}}$

3 $\dfrac{1 + 2x - x^2}{(x^2 + 1)^2}$

4 $\dfrac{2}{(1 - x)^2}$

5 $\dfrac{1}{\sqrt{2x}}$

6 0

7 $a = 2$

8 $-\dfrac{4x}{(x^2 + 1)^3}$

9 $-\dfrac{1}{4x^{3/2}}$

10 3

Beyond · very hard

1 2

Multiply by the conjugate $\sqrt{1 + 2x} + \sqrt{1 - 2x}$. The numerator becomes $(1 + 2x) - (1 - 2x) = 4x$, so the quotient is $\dfrac{4}{\sqrt{1 + 2x} + \sqrt{1 - 2x}}$, which tends to $\dfrac{4}{2} = 2$.

2 $\dfrac{2 + \sqrt{x}}{2(1 + \sqrt{x})^2}$

The quotient rule gives $\dfrac{(1 + \sqrt{x}) - x \cdot \dfrac{1}{2\sqrt{x}}}{(1 + \sqrt{x})^2}$. The numerator is $1 + \sqrt{x} - \dfrac{\sqrt{x}}{2} = 1 + \dfrac{\sqrt{x}}{2} = \dfrac{2 + \sqrt{x}}{2}$, which gives the stated result.

3 $f' = 3x^2 - 6x$, $f'' = 6x - 6$, zero at $x = 1$

Differentiating twice gives $f'(x) = 3x^2 - 6x$ and $f''(x) = 6x - 6$, which vanishes at $x = 1$. Since f'' changes sign there, $x = 1$ is a point of inflection — the subject of Topic 9.

4 Holds for negative n

Write $n = -m$ with $m > 0$, so $x^n = \dfrac{1}{x^m}$. By the reciprocal rule its derivative is $-\dfrac{(x^m)'}{x^{2m}} = -\dfrac{mx^{m-1}}{x^{2m}} = -mx^{-m-1} = nx^{n-1}$. The formula is unchanged.

5 $g'(1) = 0$

Differentiating both sides gives $f'g + fg' = 2x$. At $x = 1$: $2 \cdot 1 + 1 \cdot g'(1) = 2$, so $g'(1) = 0$.

6 $\dfrac{9}{2}$

Write $\dfrac{1-\cos 3x}{x^2} = 9 \cdot \dfrac{1-\cos 3x}{(3x)^2}$. Putting $t = 3x$, the second factor is $\dfrac{1-\cos t}{t^2}$, whose limit is $\dfrac{1}{2}$. So the whole limit is $9 \times \dfrac{1}{2} = \dfrac{9}{2}$.

7 $a \geq 0$

$\dfrac{dy}{dx} = 6x(x^2 + a)^2$, which vanishes when $x = 0$ or $x^2 = -a$. If $a > 0$ the second equation has no real root, so $x = 0$ is the only one. If $a = 0$ the derivative is $6x^5$, again zero only at $x = 0$. If $a < 0$ there are two further points $x = \pm\sqrt{-a}$. So the condition is $a \geq 0$.

8 $x = 1$ only

$\dfrac{dy}{dx} = 3x^2 - 6x + 3 = 3(x-1)^2$, which is zero only at $x = 1$. The repeated root means the gradient touches zero without changing sign, so this is a stationary point of inflection, not a turning point.

9 $\dfrac{2}{\sqrt{5}}$

$v = \dfrac{ds}{dt} = \dfrac{t}{\sqrt{t^2+1}}$. At $t = 2$ this is $\dfrac{2}{\sqrt{5}} \approx 0.894$. The velocity is always positive and less than 1, approaching 1 as t grows.

10 $\dfrac{dz}{dx} = 2f'(2x)$, $\dfrac{d^2z}{dx^2} = 4f''(2x)$

The chain rule gives $\dfrac{dz}{dx} = 2f'(2x)$. Differentiating again, the factor 2 stays and the chain rule contributes another 2, giving $4f''(2x)$. Each differentiation of a horizontal stretch brings down one more factor.

11 $c = 2$

f is a polynomial, so it is continuous on $[1, 3]$ and differentiable on $(1, 3)$. Also $f(1) = 0$ and $f(3) = 0$, so the end values agree. Then $f'(x) = 2x - 4$ is zero at $x = 2$, which lies in $(1, 3)$. The theorem is confirmed, and $x = 2$ is the vertex.

12 $\dfrac{1}{2\sqrt{2}}$

Multiply by the conjugate $\sqrt{x} + \sqrt{2}$: the numerator becomes $x - 2$, which cancels, leaving $\dfrac{1}{\sqrt{x} + \sqrt{2}}$. As $x \to 2$ this tends to $\dfrac{1}{2\sqrt{2}}$. It is the derivative of $\sqrt{x}$ at $x = 2$.

13 Proved

By the product rule $(fg)' = f'g + fg'$. Differentiating again, $(f'g)' = f''g + f'g'$ and $(fg')' = f'g' + fg''$. Adding gives $f''g + 2f'g' + fg''$. The coefficients are those of the binomial expansion, which is no accident — this is the second case of the Leibniz rule.

14 $f(x) = x^3 + x^2 + 1$

Write $f(x) = ax^3 + bx^2 + cx + d$. Then $d = f(0) = 1$ and $c = f'(0) = 0$. Also $f''(0) = 2b = 2$ gives $b = 1$, and $f'''(0) = 6a = 6$ gives $a = 1$. So $f(x) = x^3 + x^2 + 1$. The coefficients are $\dfrac{f^{(k)}(0)}{k!}$ — a Maclaurin expansion that stops.

15 $y'(0) = 100$, $y''(0) = 9900$

$y' = 100(1 + x)^{99}$, so $y'(0) = 100$. Then $y'' = 100 \times 99(1 + x)^{98}$, so $y''(0) = 9900$. These are $1!$ and $2!$ times the first two binomial coefficients after the constant term.

Topic 7 · The modulus function

Focus · easy

1 7

2 5

3 $x = 4$ or $x = -4$

4 $x = 5$ or $x = -1$

5 No solution

6 $x = 5$ or $x = -5$

7 5

8 $|x - 7|$

9 $x = -1$

10 $(3, 0)$

Practice · medium

1 $x = 4$ or $x = -3$

2 $x = 1$ or $x = -\frac{7}{3}$

3 $-3 < x < 3$

4 $x < -5$ or $x > 5$

5 $2 \le x \le 6$

6 $x < -4$ or $x > -2$

7 $(-3, 0)$

8 $(0, -2)$

9 $x = -1$

10 1

Challenge · hard

1 $x = -2$ or $x = 3$

2 $x = 3$ or $x = -\frac{1}{3}$

3 $x = 2$ or $x = -2$

4 $x = \pm 1$ and $x = \pm 2$

5 $x > \frac{1}{3}$

6 1 from the right, -1 from the left

7 $\frac{x}{|x|}$, which is 1 for $x > 0$ and -1 for $x < 0$

8 $x = 3$ or $x = -\frac{1}{3}$

9 $x = 1$ and $x = -1$

10 $x = 3$

Beyond · very hard

1 Every x with $1 \le x \le 3$

The left side is the distance from x to 1 plus the distance from x to 3. Those points are 2 apart, so for any x between them the two distances already add to exactly 2. Outside the interval the sum is larger. The solution set is the whole closed interval $[1, 3]$ — infinitely many solutions.

2 No solution

The smallest the sum of the two distances can be is the gap between the points, which is 2. Since $1 < 2$, no point of the line is close enough to both. Algebraically, each of the three cases leads to a false statement or to a root outside its own interval.

3 $x = -1$ or $x = 5$

The gap is 2, so $6 - 2 = 4$ must be shared equally outside the interval, giving 2 beyond each end: $x = 1 - 2 = -1$ or $x = 3 + 2 = 5$. Checking $x = 5$: $4 + 2 = 6$. ✓

4 Proved

For any real t, $-|t| \le t \le |t|$. Adding the two statements for a and b gives $-(|a| + |b|) \le a + b \le |a| + |b|$, which is exactly $|a + b| \le |a| + |b|$. Equality holds only when a and b have the same sign, or one is zero.

5 Proved

Write $a = (a - b) + b$ and apply the triangle inequality: $|a| \le |a - b| + |b|$, so $|a| - |b| \le |a - b|$. Swapping a and b gives $|b| - |a| \le |b - a| = |a - b|$. The modulus of a number is at most k exactly when both it and its negative are, so $||a| - |b|| \le |a - b|$.

6 $f'(x) = 2|x|$; $f''(0)$ does not exist

As in Topic 3, $f'(x) = 2|x|$ for every x, including $x = 0$. But $2|x|$ is itself a modulus function, with a corner at the origin: its one-sided derivatives there are 2 and -2. So $f''(0)$ does not exist. A function can be once differentiable everywhere without being twice differentiable.

7 $x \le 1$ or $x \ge 4$

$|t| = t$ holds exactly when $t \ge 0$. So the equation says $x^2 - 5x + 4 \ge 0$, that is $(x - 1)(x - 4) \ge 0$. A product of two factors is non-negative when both are non-negative or both non-positive, giving $x \le 1$ or $x \ge 4$.

8 2, attained at $x = 2$

The outer two terms add to at least 2, the gap between 1 and 3, with equality for every x in $[1, 3]$. Inside that interval the remaining term $|x - 2|$ is least when $x = 2$, where it is 0. So the minimum is $2 + 0 = 2$ at $x = 2$ — the median of the three points, which is the general rule for an odd number of terms.

9 Constant, equal to 2, on $0 \le x \le 2$

For $x < 0$ the sum is $-x + (2 - x) = 2 - 2x$, falling. For $0 \le x \le 2$ it is $x + (2 - x) = 2$, constant. For $x > 2$ it is $x + (x - 2) = 2x - 2$, rising. The graph is a flat-bottomed valley with a horizontal floor of length 2, not a single vertex.

10 $x \ge 1$

The statement $|t| = t$ holds exactly when $t \ge 0$, so $x - 1 \ge 0$ and $x \ge 1$. Every such x works, so the answer is a half-line, not a point.

11 $-2 \le x \le 2$

The left side is the distance from x to 1 plus the distance to -1. Those points are 2 apart, so the sum is 2 throughout $[-1, 1]$ and grows by 2 for each unit outside. To reach 4 requires 1 unit beyond either end, so the sum is at most 4 exactly on $[-2, 2]$.

12 $\sqrt{x^2} = |x|$

The radical sign denotes the non-negative square root. Squaring destroys the sign of x, and the root cannot restore it, so the output is always non-negative. For $x = -3$, $\sqrt{(-3)^2} = \sqrt{9} = 3 = |-3|$, not -3. The identity $\sqrt{x^2} = x$ is true only for $x \ge 0$.

13 $a = -2$ or $a = 4$

Substituting gives $|1 - a| = 3$, so $1 - a = 3$ or $1 - a = -3$, that is $a = -2$ or $a = 4$. Read as distance: a must be 3 away from 1, and there are two such points, one on each side.

14 Continuous, not differentiable

Continuity: both one-sided limits at 0 equal 0, which is $|0|$, so the function is continuous. Differentiability: the quotient $\dfrac{|h|}{h}$ is 1 for $h > 0$ and -1 for $h < 0$, so the one-sided derivatives are 1 and -1. They disagree, so no derivative exists. This is the standard example showing continuity does not imply differentiability.

15 $x > \dfrac{1}{2}$

Both sides are non-negative, so squaring is safe: $x^2 - 4x + 4 < x^2 + 2x + 1$, giving $-6x < -3$ and $x > \dfrac{1}{2}$. As distance: x must be nearer to 2 than to -1, so it lies on the far side of the midpoint $\dfrac{1}{2}$.

Topic 8 · The equations of the tangent and the normal

Focus · easy

1 4

2 $y = 4x - 4$

6 $x = 0$

7 -2

3 $-\frac{1}{4}$

4 3

5 $y = 3x + 1$ — the line is its own tangent

8 $-\frac{1}{2}$

9 -1

10 $x = 0$

Practice · medium

1 $y = 2x$

2 $y = 2x - 2$

3 $y = \frac{x}{4} + 1$

4 $y = -x + 2$

5 $y = -\frac{x}{2} + \frac{3}{2}$

6 $y = -\frac{x}{2} + 1$

7 $x = 1$ and $x = -1$

8 $x = -\frac{3}{2}$

9 $y = 4x - 2$

10 -9

Challenge · hard

1 $a = 2$ or $a = -2$

2 $y = -\frac{x}{4} + 1$

3 $y = 2$ and $y = -2$

4 $y = -2x + 3$

5 $x = -\frac{1}{2}$

6 $x = \frac{5}{2}$

7 -2

8 $(-2, -8)$

9 2

10 $b = 3$

Beyond · very hard

1 $y = 2x - 1$ and $y = -2x - 1$

The tangent at $x = a$ is $y = 2ax - a^2$. Putting $(0, -1)$ into it gives $-1 = -a^2$, so $a = \pm 1$. Substituting back gives $y = 2x - 1$ and $y = -2x - 1$. A point below the parabola always has two tangents through it; a point inside has none.

2 $(-\frac{3}{2}, \frac{9}{4})$

The tangent gradient at $x = 1$ is 2, so the normal gradient is $-\frac{1}{2}$ and the normal is $y = -\frac{x}{2} + \frac{3}{2}$. Setting $x^2 = -\frac{x}{2} + \frac{3}{2}$ gives $2x^2 + x - 3 = 0$, that is $(2x + 3)(x - 1) = 0$. The new root is $x = -\frac{3}{2}$, giving $y = \frac{9}{4}$.

3 The area is always 2

The gradient is $-\frac{1}{a^2}$, so the tangent is $y = -\frac{x}{a^2} + \frac{2}{a}$. Its x-intercept is $2a$ and its y-intercept is $\frac{2}{a}$. The area is $\frac{1}{2} \times 2a \times \frac{2}{a} = 2$, free of a. Every tangent to this hyperbola cuts off the same area.

4 The point of tangency bisects AB

From the previous question $A = (2a, 0)$ and $B = (0, \frac{2}{a})$. Their midpoint is $(a, \frac{1}{a})$, which is the point of tangency. This bisection property, like the constant area, belongs to the rectangular hyperbola alone.

5 $c = -\frac{1}{4}$

Setting $x^2 = x + c$ gives $x^2 - x - c = 0$. A tangent touches, so this must have a repeated root and the discriminant must vanish: $1 + 4c = 0$, giving $c = -\frac{1}{4}$. The point of contact is $x = \frac{1}{2}$.

6 $m = 2$ and $m = -2$

Setting $x^2 = mx - 1$ gives $x^2 - mx + 1 = 0$, whose discriminant is $m^2 - 4$. For a repeated root this must be zero, so $m = \pm 2$. These are the same two tangents as in question 1, which is no surprise: both lines pass through $(0, -1)$.

7 $\frac{5}{2}$

The tangent $y = 2x - 1$ meets the axis at $P(0, -1)$; the normal $y = -\frac{x}{2} + \frac{3}{2}$ meets it at $Q(0, \frac{3}{2})$. The distance is $\frac{3}{2} - (-1) = \frac{5}{2}$.

8 It meets the curve again at $x = -2a$

The tangent is $y = 3a^2x - 2a^3$. Setting $x^3 = 3a^2x - 2a^3$ gives $x^3 - 3a^2x + 2a^3 = 0$, which factorises as $(x - a)^2(x + 2a) = 0$. The double root $x = a$ is the point of contact, and the remaining root is $x = -2a$. Only $a = 0$ makes them coincide.

9 $(2, 2)$ and $(-2, -2)$

Parallel means equal gradients, so $3x^2 - 3 = 9$, giving $x^2 = 4$ and $x = \pm 2$. Then $y = 8 - 6 = 2$ and $y = -8 + 6 = -2$, so the points are $(2, 2)$ and $(-2, -2)$.

10 $\arctan\frac{3}{4} \approx 36.9°$

The gradients are $2x = 2$ and $\frac{1}{2\sqrt{x}} = \frac{1}{2}$. The angle between two lines of gradients m_1 and m_2 satisfies $\tan\theta = |\frac{m_1 - m_2}{1 + m_1 m_2}| = |\frac{2 - 0.5}{1 + 1}| = \frac{3}{4}$, so $\theta = \arctan\frac{3}{4} \approx 36.9°$. The curves are reflections of each other in $y = x$, so the line $y = x$ bisects this angle.

11 No — the gradients give a product of -2

They meet where $x^2 = \frac{1}{x}$, that is $x^3 = 1$ and $x = 1$, at the point $(1, 1)$. The gradients there are 2 and -1, whose product is -2, not -1. So the curves cross at an angle, not perpendicularly.

12 $k = 1$

Setting $x^2 + k = 2x$ gives $x^2 - 2x + k = 0$, with discriminant $4 - 4k$. Touching means a repeated root, so $4 - 4k = 0$ and $k = 1$. The contact point is $x = 1$, that is $(1, 2)$.

13 $b = a^2 + \frac{1}{2}$

The normal gradient is $-\frac{1}{2a}$, so the normal is $y - a^2 = -\frac{1}{2a}(x - a)$. Putting $x = 0$ gives $y = a^2 + \frac{1}{2}$. Every normal cuts the axis exactly $\frac{1}{2}$ above the height of its own point of contact.

14 $x = -a - \dfrac{1}{2a}$

Setting $x^2 = -\dfrac{1}{2a}(x - a) + a^2$ and multiplying by $2a$ gives $2ax^2 + x - 2a^3 - a = 0$. One root is $x = a$, as it must be. Since the roots multiply to $\dfrac{-2a^3 - a}{2a} = -a^2 - \dfrac{1}{2}$, the other root is $\dfrac{-a^2 - 1/2}{a} = -a - \dfrac{1}{2a}$.

15 $(-1, -1)$

The tangent at $x = a$ is $y = 3a^2x - 2a^3$. Putting $(0, 2)$ into it gives $2 = -2a^3$, so $a^3 = -1$ and $a = -1$, the only real root. The point is $(-1, -1)$ and the tangent is $y = 3x + 2$. Unlike the parabola, the cubic gives just one such tangent here.

Topic 9 · Investigating a function with the derivative

Focus · easy

1 $(2, -4)$

2 Minimum

3 $3x^2 - 3$

4 $x = 1$ and $x = -1$

5 $x > 0$

6 $x > 0$

7 2

8 $6x$

9 $x = 0$

10 Maximum

Practice · medium

1 $x = 0$ and $x = 2$

2 Maximum at $x = 0$, minimum at $x = 2$

3 $x = 1$

4 $x = 1$ and $x = 2$

5 $x < -2$ or $x > 2$

6 $x = 0$, $x = 1$ and $x = -1$

7 $x = 1$ and $x = -1$

8 Minimum

9 $3x^2 - 6x$

10 9

Challenge · hard

1 Maximum at $x = 1$, minimum at $x = 3$

2 $x = 2$

3 $x = 1$ and $x = -1$

4 $x = 0$ and $x = 3$

5 A stationary point of inflection

6 Greatest 2, least -2

7 Increasing on each branch

8 $\dfrac{dy}{dx} = 3x^2 + 1 > 0$ always

9 Minimum at $x = -1$

10 $a > 0$

Beyond · very hard

1 $a = -6$, $b = 9$

$\dfrac{dy}{dx} = 3x^2 + 2ax + b$ must have roots 1 and 3. Their sum is $4 = -\dfrac{2a}{3}$, giving $a = -6$; their product is $3 = \dfrac{b}{3}$, giving $b = 9$. The curve is $y = x^3 - 6x^2 + 9x$, the one met in the Challenge set.

2 Two, one repeated, or none

For $y = ax^3 + bx^2 + cx + d$ with $a \neq 0$, $\dfrac{dy}{dx} = 3ax^2 + 2bx + c$ is a genuine quadratic. A quadratic has two distinct roots, one repeated root, or no real roots, according to the sign of its discriminant $4b^2 - 12ac$. A cubic can therefore never have exactly three turning points, however it is drawn.

3 Minimum at $x = 3$; stationary inflection at $x = 0$

$\frac{dy}{dx} = 4x^3 - 12x^2 = 4x^2(x - 3)$, zero at $x = 0$ and $x = 3$. Then $\frac{d^2y}{dx^2} = 12x^2 - 24x$, which is $36 > 0$ at $x = 3$: a minimum. At $x = 0$ it is 0, so the test fails; but $4x^2(x - 3)$ is negative on both sides of 0, so there is no sign change and $x = 0$ is a stationary point of inflection.

4 It is monotonic there

Let $p < q$ be consecutive stationary points, so f' has no zero in (p, q). If f' took both a positive and a negative value there, then by the intermediate value theorem applied to f' it would have to vanish somewhere between — contradicting that p and q are consecutive. So f' keeps one sign throughout, and f is increasing or decreasing across the whole interval.

5 2, at $x = 1$

$\frac{dy}{dx} = 1 - \frac{1}{x^2}$ is zero at $x = 1$ in this range. Then $\frac{d^2y}{dx^2} = \frac{2}{x^3} = 2 > 0$, so it is a minimum, with value $1 + 1 = 2$. As $x \to 0^+$ or $x \to \infty$ the function grows without bound, so this local minimum is the global one.

6 Equality only at $x = 1$

The global minimum of $x + \frac{1}{x}$ on $x > 0$ is 2, attained only at $x = 1$, so the inequality holds everywhere on that range with equality only there. Replacing x by $\frac{a}{b}$ gives $\frac{a}{b} + \frac{b}{a} \geq 2$, which rearranges to $a^2 + b^2 \geq 2ab$ — the arithmetic–geometric mean inequality for two numbers.

7 $x = -\frac{b}{3a}$

$\frac{d^2y}{dx^2} = 6ax + 2b$, which vanishes at $x = -\frac{b}{3a}$. Since $6ax + 2b$ is linear with non-zero gradient $6a$, it changes sign there, so the point really is an inflection. It is also the mean of the two stationary points when they exist, because they are the roots of a quadratic whose axis of symmetry is this same value.

8 4, at $x = 2$

$\frac{dy}{dx} = 6x - 3x^2 = 3x(2 - x)$, zero at $x = 0$ and $x = 2$. The values are $f(0) = 0$, $f(2) = 4 \cdot 1 = 4$ and $f(3) = 0$ at the far endpoint. The greatest is 4.

9 Greatest $\frac{1}{2}$ at $x = 1$, least $-\frac{1}{2}$ at $x = -1$

$\frac{dy}{dx} = \frac{1 - x^2}{(x^2 + 1)^2}$, zero at $x = \pm 1$. The values are $\frac{1}{2}$ and $-\frac{1}{2}$. As $x \to \pm\infty$ the function tends to 0, so these local extremes are global. The function is odd, which is why the two values are opposite.

10 A stationary inflection at $x = 1$

$f'(x) = 3x^2 - 6x + 3 = 3(x - 1)^2$, which is zero only at $x = 1$ and never negative. So f is increasing everywhere, pausing at $x = 1$ without turning. Indeed $f(x) = (x - 1)^3$, the cubic curve translated one unit to the right.

11 $4p^3 + 27q^2 = 0$

A stationary point on the axis is a repeated root of the cubic. At a stationary point $3x^2 + p = 0$, so $x^2 = -\frac{p}{3}$ (which needs $p \leq 0$) and $x^3 = -\frac{px}{3}$. Substituting into $x^3 + px + q = 0$ gives $\frac{2px}{3} + q = 0$, so $x = -\frac{3q}{2p}$. Squaring and matching with $x^2 = -\frac{p}{3}$ gives $\frac{9q^2}{4p^2} = -\frac{p}{3}$, that is $27q^2 = -4p^3$.

12 At most one

$f'' > 0$ means f' is strictly increasing. A strictly increasing function takes each value at most once, so $f'(x) = 0$ has at most one solution. If it has one, the point is a minimum, since f' goes from negative to positive there.

13 Minimum 2 at $x = 1$; maximum -2 at $x = -1$

Simplify first: $y = x + \frac{1}{x}$. Then $\frac{dy}{dx} = 1 - \frac{1}{x^2}$ vanishes at $x = \pm 1$, and $\frac{d^2y}{dx^2} = \frac{2}{x^3}$ is $2 > 0$ at $x = 1$ and $-2 < 0$ at $x = -1$. Note the local maximum -2 lies below the local minimum 2 — possible only because the two branches are separated by the asymptote $x = 0$.

14 Three if $-2 < k < 2$; two if $k = \pm 2$; one otherwise

Let $f(x) = x^3 - 3x + k$. Then $f'(x) = 3x^2 - 3$ gives stationary points at $x = \pm 1$, with $f(-1) = 2 + k$ a local maximum and $f(1) = -2 + k$ a local minimum. The curve crosses the axis three times exactly when these have opposite signs, that is $(2 + k)(k - 2) < 0$, so $-2 < k < 2$. When $k = \pm 2$ one of them sits on the axis, giving a repeated root and two distinct roots in all. Otherwise there is one.

15 At most $n - 1$

If f has degree n then f' has degree exactly $n - 1$, since the leading term $a_n x^n$ differentiates to $n a_n x^{n-1}$ with $n a_n \neq 0$. A non-zero polynomial of degree $n - 1$ has at most $n - 1$ real roots, and the stationary points are precisely those roots. So there are at most $n - 1$ of them — which is why a quadratic has one turning point and a cubic at most two.

Topic 10 · Extremum problems

Focus · easy

1 25

2 25

3 25 cm²

4 4

5 9

6 -9

7 $x = -\frac{1}{2}$

8 3

9 16

10 4

Practice · medium

1 72

2 81 cm²

3 12

4 32 cm

5 6

6 9

7 12

8 $x = 4$ cm

9 1250 m²

10 2

Challenge · hard

1 128 cm³

2 r^2

3 $\frac{32\sqrt{3}}{9}$

4 $r = (\frac{500}{\pi})^{1/3} \approx 5.42$ cm

5 $x = \pm\sqrt{\frac{5}{2}}$

6 $\frac{400}{\pi + 4} \approx 56.0$ cm

7 2, at $x = 1$

8 7200 m²

9 $\frac{4a^3}{27}$, at $x = \frac{a}{3}$

10 $(-\frac{6}{5}, \frac{3}{5})$

Beyond · very hard

1 $\frac{32}{81}\pi R^3$

Let the cone have height h and base radius r, with its apex on the sphere. The base circle lies at distance $h - R$ from the centre, so $r^2 = R^2 - (h-R)^2 = 2Rh - h^2$. Then $V = \frac{1}{3}\pi r^2 h = \frac{\pi}{3}(2Rh^2 - h^3)$. Setting $\frac{dV}{dh} = \frac{\pi}{3}(4Rh - 3h^2) = 0$ gives $h = \frac{4R}{3}$. Then $r^2 = \frac{8R^2}{9}$ and $V = \frac{\pi}{3}\cdot\frac{8R^2}{9}\cdot\frac{4R}{3} = \frac{32\pi R^3}{81}$.

2 $\frac{4\pi R^3}{3\sqrt{3}}$

With base radius r and height h, the corner of the cylinder lies on the sphere, so $r^2 + (\frac{h}{2})^2 = R^2$. Then $V = \pi r^2 h = \pi(R^2 - \frac{h^2}{4})h$. Setting $\frac{dV}{dh} = \pi(R^2 - \frac{3h^2}{4}) = 0$ gives $h = \frac{2R}{\sqrt{3}}$ and $r^2 = \frac{2R^2}{3}$, so $V = \frac{4\pi R^3}{3\sqrt{3}}$.

3 $h = r = (\frac{V}{\pi})^{1/3}$

$S = \pi r^2 + 2\pi r h$ with $V = \pi r^2 h$, so $h = \frac{V}{\pi r^2}$ and $S = \pi r^2 + \frac{2V}{r}$. Then $\frac{dS}{dr} = 2\pi r - \frac{2V}{r^2} = 0$ gives $r^3 = \frac{V}{\pi}$. Substituting back, $h = \frac{V}{\pi r^2} = \frac{r^3}{r^2} = r$. Since $\frac{d^2S}{dr^2} = 2\pi + \frac{4V}{r^3} > 0$, it is a minimum.

4 $(a^{2/3} + b^{2/3})^{3/2}$

A ladder touching both walls and the corner at angle θ to one corridor has length $L(\theta) = \frac{a}{\sin\theta} + \frac{b}{\cos\theta}$. The longest ladder that fits is the shortest such L. Setting $L'(\theta) = -\frac{a\cos\theta}{\sin^2\theta} + \frac{b\sin\theta}{\cos^2\theta} = 0$ gives $\tan^3\theta = \frac{a}{b}$. Substituting back and simplifying gives $L = (a^{2/3} + b^{2/3})^{3/2}$.

5 The law of refraction

Put the boundary on the x-axis, the source at height h_1 above a point O and the target at depth h_2 below a point d along. If the ray crosses at x, the time is $T(x) = \frac{\sqrt{h_1^2 + x^2}}{v_1} + \frac{\sqrt{h_2^2 + (d-x)^2}}{v_2}$. Then $T'(x) = \frac{x}{v_1\sqrt{h_1^2 + x^2}} - \frac{d-x}{v_2\sqrt{h_2^2 + (d-x)^2}}$. The two quotients are exactly $\frac{\sin\theta_1}{v_1}$ and $\frac{\sin\theta_2}{v_2}$, so $T' = 0$ gives the stated law.

6 $\frac{ab}{4}$

Put the right angle at the origin with legs along the axes; the hypotenuse is $\frac{x}{a} + \frac{y}{b} = 1$. A rectangle with its far corner on that line has area $A = xy = x\cdot b(1 - \frac{x}{a}) = bx - \frac{b}{a}x^2$. Then $A' = b - \frac{2b}{a}x = 0$ gives $x = \frac{a}{2}$, so $y = \frac{b}{2}$ and $A = \frac{ab}{4}$ — exactly half the triangle.

7 $x = \pm\sqrt{\frac{3}{2}}$; distance $\frac{\sqrt{7}}{2}$

Minimise $d^2 = x^2 + (x^2 - 2)^2$. Writing $u = x^2 \ge 0$ gives $d^2 = u + (u-2)^2 = u^2 - 3u + 4$, a quadratic least at $u = \frac{3}{2}$. So $x^2 = \frac{3}{2}$ and $d^2 = \frac{9}{4} - \frac{9}{2} + 4 = \frac{7}{4}$, giving $d = \frac{\sqrt{7}}{2} \approx 1.32$. Two points are equally near, by symmetry.

8 $\frac{108}{3125}$, at $x = \frac{2}{5}$

$f'(x) = 2x(1-x)^3 - 3x^2(1-x)^2 = x(1-x)^2[2(1-x) - 3x] = x(1-x)^2(2-5x)$. In $(0, 1)$ this vanishes only at $x = \frac{2}{5}$, and f is zero at both endpoints, so that is the maximum. Its value is $(\frac{2}{5})^2(\frac{3}{5})^3 = \frac{4}{25} \cdot \frac{27}{125} = \frac{108}{3125}$.

9 $x = (2V)^{1/3}$

$S = x^2 + 4xh$ with $x^2h = V$, so $h = \frac{V}{x^2}$ and $S = x^2 + \frac{4V}{x}$. Then $\frac{dS}{dx} = 2x - \frac{4V}{x^2} = 0$ gives $x^3 = 2V$, so $x = (2V)^{1/3}$. The height is then $h = \frac{V}{x^2} = \frac{x}{2}$ — the box is twice as wide as it is tall.

10 $\frac{1}{2}ab$, with a right angle between them

The area is $\frac{1}{2}ab \cdot \sin\theta$, where θ is the angle between the two sides. Differentiating with respect to θ gives $\frac{1}{2}ab \cdot \cos\theta$, which vanishes at $\theta = \frac{\pi}{2}$, where the second derivative $-\frac{1}{2}ab \cdot \sin\theta$ is negative. So the area is greatest at a right angle, with value $\frac{1}{2}ab$.

11 Proved, with equality only when $a = b$

Fix the product $ab = p$ and let $a = x$, so $b = \frac{p}{x}$ with $x > 0$. The sum is $S(x) = x + \frac{p}{x}$, with $S'(x) = 1 - \frac{p}{x^2}$, zero at $x = \sqrt{p}$, and $S'' = \frac{2p}{x^3} > 0$, so that is a minimum. The least sum is $\sqrt{p} + \sqrt{p} = 2\sqrt{p} = 2\sqrt{ab}$, so $a + b \geq 2\sqrt{ab}$ always, with equality exactly when $a = b = \sqrt{p}$.

12 Width $\frac{d}{\sqrt{3}}$, depth $d\sqrt{\frac{2}{3}}$

With width w and depth h the diagonal is the diameter, so $w^2 + h^2 = d^2$. Strength $S = wh^2 = w(d^2 - w^2) = d^2w - w^3$. Then $\frac{dS}{dw} = d^2 - 3w^2 = 0$ gives $w = \frac{d}{\sqrt{3}}$, and $h^2 = d^2 - \frac{d^2}{3} = \frac{2d^2}{3}$. The depth is $\sqrt{2}$ times the width.

13 The square is greatest

With perimeter P fixed, write the sides as x and $\frac{P}{2} - x$ for $0 < x < \frac{P}{2}$. The area is $A(x) = x(\frac{P}{2} - x) = \frac{Px}{2} - x^2$, so $A'(x) = \frac{P}{2} - 2x$, zero at $x = \frac{P}{4}$, and $A'' = -2 < 0$, a maximum. Both sides are then $\frac{P}{4}$, so the rectangle is a square, of area $\frac{P^2}{16}$.

14 Radius $\frac{2R}{3}$, height $\frac{H}{3}$; volume $\frac{4}{27}\pi R^2 H$

A cylinder of radius r inscribed in the cone has height $h = H(1 - \frac{r}{R})$, by similar triangles. Then $V = \pi r^2 h = \pi H(r^2 - \frac{r^3}{R})$. Setting $\frac{dV}{dr} = \pi H(2r - \frac{3r^2}{R}) = 0$ gives $r = \frac{2R}{3}$, so $h = \frac{H}{3}$ and $V = \pi \cdot \frac{4R^2}{9} \cdot \frac{H}{3} = \frac{4\pi R^2 H}{27}$ — four ninths of the cone.

15 $\dfrac{50}{\pi + 4} \approx 7.0$ m²

Let the semicircle have radius r, so the rectangle is $2r$ wide and h tall. The perimeter is $2r + 2h + \pi r = 10$, giving $h = \dfrac{10 - 2r - \pi r}{2}$. The area is $A = 2rh + \dfrac{\pi r^2}{2} = 10r - 2r^2 - \dfrac{\pi r^2}{2}$. Then $\dfrac{dA}{dr} = 10 - 4r - \pi r = 0$ gives $r = \dfrac{10}{\pi + 4}$, and substituting back gives $A = 5r = \dfrac{50}{\pi + 4} \approx 7.0$ m². At that radius $h = r$, so the rectangle is twice as wide as it is tall.

Topic 11 · Revision · Control work 2 — applications of the derivative

Focus · easy

1 $y = 2x - 1$

2 $x = 1$

3 Minimum

4 $3x^2 - 12$

5 10

6 $-\dfrac{1}{3}$

7 $x < 0$

8 16

9 2

10 1

Practice · medium

1 $x = 1$ and $x = 3$

2 Maximum at $x = 1$, minimum at $x = 3$

3 $x = \dfrac{3}{2}$

4 $y = -2x + 4$

5 100 cm²

6 4

7 $x = 0$, $x = \sqrt{2}$ and $x = -\sqrt{2}$

8 $y = -4x + 18$

9 $x < 0$ or $x > 2$

10 4

Challenge · hard

1 Maximum at $x = 0$, minimum at $x = 2$

2 Greatest 18, least -2

3 $a = 0$ or $a = 2$

4 432 cm³

5 -3

6 $a = -3$

7 $r = (\dfrac{250}{\pi})^{1/3} \approx 4.30$ cm

8 $\dfrac{dy}{dx} = \dfrac{1}{(x + 1)^2} > 0$

9 16

10 32, at $x = 4$

Beyond · very hard

1 $a = -3$, $b = -9$, $c = 5$

$\dfrac{dy}{dx} = 3x^2 + 2ax + b$ must have roots -1 and 3. Their sum is $2 = -\dfrac{2a}{3}$, so $a = -3$; their product is $-3 = \dfrac{b}{3}$, so $b = -9$. Passing through $(0, 5)$ gives $c = 5$. Check: $\dfrac{d^2y}{dx^2} = 6x - 6$ is $-12 < 0$ at $x = -1$ and $12 > 0$ at $x = 3$, as required.

2 Three if $-2 < k < 2$; two if $k = \pm 2$; one otherwise

The curve $y = x^3 - 3x$ has a local maximum 2 at $x = -1$ and a local minimum -2 at $x = 1$. A horizontal line $y = k$ cuts it three times when it passes strictly between those heights, twice when it passes through one of them, and once otherwise.

3 $\sqrt{2}$, at the point $(-1, 1)$

Minimise $d^2 = x^2 + (x + 2)^2 = 2x^2 + 4x + 4$. Then $\frac{d}{dx}(d^2) = 4x + 4 = 0$ gives $x = -1$, with second derivative $4 > 0$. So $y = 1$ and $d^2 = 1 + 1 = 2$, giving $d = \sqrt{2}$. The segment from the origin to $(-1, 1)$ has gradient -1, perpendicular to the line, as it must be.

4 $\dfrac{a(4a^2 + 1)}{4}$

The tangent $y = 2ax - a^2$ meets the axis at $(0, -a^2)$; the normal $y = -\frac{x}{2a} + a^2 + \frac{1}{2}$ meets it at $(0, a^2 + \frac{1}{2})$. The base along the axis has length $(a^2 + \frac{1}{2}) + a^2 = 2a^2 + \frac{1}{2}$, and the height is a. The area is $\frac{1}{2}a(2a^2 + \frac{1}{2}) = \dfrac{a(4a^2 + 1)}{4}$.

5 $2ab$

With a corner at (x, y) in the first quadrant the area is $A = 4xy = 4bx\sqrt{1 - \frac{x^2}{a^2}}$. Maximising $A^2 = 16b^2(x^2 - \frac{x^4}{a^2})$ gives $2x - \frac{4x^3}{a^2} = 0$, so $x = \frac{a}{\sqrt{2}}$ and then $y = \frac{b}{\sqrt{2}}$. The area is $4 \cdot \frac{a}{\sqrt{2}} \cdot \frac{b}{\sqrt{2}} = 2ab$ — exactly half the area of the bounding rectangle $2a \times 2b$.

6 The square, of area $\dfrac{d^2}{2}$

With sides w and h and diagonal d, $w^2 + h^2 = d^2$. Maximise $A^2 = w^2h^2 = w^2(d^2 - w^2)$. Writing $u = w^2$, this is $u(d^2 - u)$, a downward parabola in u greatest at $u = \frac{d^2}{2}$. Then $w^2 = h^2 = \frac{d^2}{2}$, so $w = h$ and the area is $\frac{d^2}{2}$.

7 Increasing for all x iff $p \geq 0$

$f'(x) = 3x^2 + p$. If $p \geq 0$ this is non-negative everywhere, and zero at most at the single point $x = 0$, so f increases throughout. If $p < 0$ then f' is negative for $|x| < \sqrt{-\frac{p}{3}}$, so f decreases on that interval. Hence the condition is exactly $p \geq 0$ — and it matches the discriminant test of Topic 9.

8 Maximum at $x = 0$; minima at $x = 2$ and $x = -2$

$\frac{dy}{dx} = 4x^3 - 16x = 4x(x^2 - 4)$, zero at $x = 0$ and $x = \pm 2$. Then $\frac{d^2y}{dx^2} = 12x^2 - 16$: at $x = 0$ it is $-16 < 0$, a maximum of 0; at $x = \pm 2$ it is $32 > 0$, minima of $16 - 32 = -16$. The curve is even, which is why the two minima match.

9 12

$v = 3t^2 - 12t + 9 = 3(t - 1)(t - 3)$, so the particle is at rest at $t = 1$ and $t = 3$. The positions are $s(0) = 0$, $s(1) = 4$, $s(3) = 0$ and $s(4) = 4$. Distance is the sum of the moduli of the changes: $4 + 4 + 4 = 12$. The displacement is only 4 — the difference is exactly why the journey has to be split at the instants of rest.

10 The inflection is their mean

For $y = ax^3 + bx^2 + cx + d$, the stationary points are the roots of $3ax^2 + 2bx + c = 0$, whose sum is $-\frac{2b}{3a}$, so their mean is $-\frac{b}{3a}$. The inflection solves $6ax + 2b = 0$, giving the same $x = -\frac{b}{3a}$. Every cubic is therefore symmetric about its point of inflection.

11 $\frac{3\sqrt{3}}{4}R^2$, for the equilateral triangle

Fix a chord as base. For a given base the area is greatest when the apex is as far from it as possible, which puts the apex on the perpendicular bisector — so the triangle is isosceles. Writing the apex angle at the centre and maximising, or testing the equilateral case, gives side $s = R\sqrt{3}$ and area $\frac{\sqrt{3}}{4}s^2 = \frac{\sqrt{3}}{4}\cdot 3R^2 = \frac{3\sqrt{3}}{4}R^2 \approx 1.30R^2$.

12 Increasing everywhere

$f'(x) = 1 + \cos x$. Since $\cos x \geq -1$, this is never negative. It equals zero only when $\cos x = -1$, that is at the isolated points $x = \pi + 2k\pi$. A function whose derivative is non-negative, and zero only at isolated points, is strictly increasing, so f never turns back — it merely flattens momentarily at each of those points.

13 $a = -1$, $b = 3$

$\frac{dy}{dx} = 3ax^2 + 2bx$, and at $x = 2$ this gives $12a + 4b = 0$, so $b = -3a$. Passing through $(2, 4)$ gives $8a + 4b = 4$, and substituting $b = -3a$ makes it $8a - 12a = -4a = 4$, so $a = -1$ and $b = 3$. The curve is $y = -x^3 + 3x^2$.

14 $h = 2r$

With S fixed, $S = 2\pi r^2 + 2\pi rh$ gives $h = \frac{S - 2\pi r^2}{2\pi r}$, so $V = \pi r^2 h = \frac{Sr - 2\pi r^3}{2}$. Then $\frac{dV}{dr} = \frac{S - 6\pi r^2}{2} = 0$ gives $S = 6\pi r^2$. Substituting back, $h = \frac{6\pi r^2 - 2\pi r^2}{2\pi r} = 2r$. Since $\frac{d^2V}{dr^2} = -6\pi r < 0$, it is a maximum.

15 They meet at $(\frac{a + b}{2}, ab)$

The two tangents are $y = 2ax - a^2$ and $y = 2bx - b^2$. Equating gives $2x(a - b) = a^2 - b^2 = (a - b)(a + b)$, and since $a \neq b$ we may divide to get $x = \frac{a + b}{2}$. Substituting into the first tangent gives $y = a(a + b) - a^2 = ab$. So the meeting point sits above the midpoint of the two x-values, at the height of their product.

Topic 12 · Approximate calculations with the derivative

Focus · easy

1 0.06

2 1.2

3 $\frac{1}{60} \approx 0.0167$

4 10.05

5 2.005

6 4.04

7 1.006

8 -0.025

9 0.99

10 $\Delta y \approx f'(x) \cdot \Delta x$

Practice · medium

1 5.1

2 2.0083

3 27.54

4 0.4975

5 2 cm²

6 2 %

7 $0.5\pi \approx 1.57$ cm²

8 2.16 cm³

9 0.02

10 0.01

Challenge · hard

1 10.100

6 3 %

2 10.0033

3 0.515

4 0.05

5 1.02

7 4 %

8 16.32

9 $1 + nx$

10 0.99

Beyond · very hard

1 Of order $(\Delta x)^2$

Expand $f(a + h)$ as a Taylor series: $f(a) + f'(a)h + \frac{f''(a)}{2}h^2 + \dots$. The linear estimate keeps the first two terms, so the error is $\frac{f''(a)}{2}h^2$ plus higher powers. Halving the step therefore cuts the error by a factor of about four — which is why the method is good for $\sqrt{101}$ and poor for $\sqrt{200}$.

2 7.0714; the true value is 7.07107, so the estimate is high by about 0.0004

Take $f(x) = \sqrt{x}$ at $x = 49$, where $f = 7$ and $f' = \frac{1}{14}$. Then $\sqrt{50} \approx 7 + \frac{1}{14} = 7.0714$. The error agrees with the previous question: $f'' = -\frac{1}{4x^{3/2}}$, so $\frac{|f''|}{2}h^2 \approx \frac{1}{8(343)} \approx 0.0004$.

3 $1 + \frac{x}{2}$; it gives 1.05 against the true 1.0488

With $f(x) = \sqrt{1 + x}$, $f(0) = 1$ and $f'(x) = \frac{1}{2\sqrt{1 + x}}$, so $f'(0) = \frac{1}{2}$. The tangent at the origin is $y = 1 + \frac{x}{2}$. At $x = 0.1$ it gives 1.05, against $\sqrt{1.1} = 1.0488$ — high by about 0.0012, as the curve is concave down.

4 The estimate is 2; the true value is 2.5937

With $f(x) = x^{10}$ at $x = 1$, $f' = 10$, so $(1.1)^{10} \approx 1 + 10(0.1) = 2$. The true value is 2.5937, so the estimate is out by 23 %. A step of 0.1 is not small when the tenth derivative is large: the method is only as good as the step is short.

5 0.5 %

T is proportional to $L^{1/2}$, and the relative error in L^n is n times the relative error in L. With $n = \frac{1}{2}$ the period lengthens by about 0.5 %. To slow a clock by 1 % the pendulum must be lengthened by 2 %.

6 6 %

$V = \frac{1}{3}\pi r^2 h$ with h fixed, so V is proportional to r^2 and the relative change doubles: about 6 %. Had the height varied instead, with r fixed, the change would have been 3 %, since V is only linear in h.

7 Always an underestimate

$f''(x) = \frac{2}{x^3} > 0$ for $x > 0$, so the curve is concave up and lies above every one of its tangents. The linear estimate reads off the tangent, so it falls short. The same argument makes the estimate of $\sqrt{x}$ an overestimate, since that curve is concave down.

8 1.4142

Here $f'(x) = 2x$, so the iteration is $x_{n+1} = \frac{1}{2}(x_n + \frac{2}{x_n})$. From $x_0 = 1.5$: $x_1 = 1.41667$, $x_2 = 1.414216$, $x_3 = 1.4142136$. The digits roughly double each step, which is what makes the method worth the extra derivative.

9 1.3247

$f'(x) = 3x^2 - 1$. From $x_0 = 1.5$: $f = 0.875$ and $f' = 5.75$, so $x_1 = 1.3478$. Then $f = 0.1007$ and $f' = 4.4501$, giving $x_2 = 1.3252$, and $x_3 = 1.32472$. This number is the plastic ratio, the real root of the equation.

10 The factor is n

With $y = x^n$, $\dfrac{dy}{dx} = nx^{n-1}$, so $\Delta y \approx nx^{n-1}\Delta x$. Dividing by $y = x^n$ gives $\dfrac{\Delta y}{y} \approx n\dfrac{\Delta x}{x}$. The result holds for any real n, so a 1 % error in a radius gives 2 % in an area, 3 % in a volume and 0.5 % in a square root.

11 0.5151

Work from $60° = \dfrac{\pi}{3}$, where $\cos = 0.5$ and the derivative is $-\sin 60° = -0.8660$. The step is $-1° = -0.01745$ radians, so $\cos 59° \approx 0.5 - 0.8660(-0.01745) = 0.5 + 0.0151 = 0.5151$. The true value is 0.51504. The step must be in radians even though the angle is quoted in degrees.

12 1.00995, against the linear 1.01

The linear estimate is $1 + 0.01 = 1.01$. The next term is $-\dfrac{(0.02)^2}{8} = -0.00005$, giving 1.00995. The true value is 1.0099505, so the quadratic term has cut the error from 5×10^{-5} to under 10^{-7}.

13 2 %

Here $\dfrac{1}{x^2} = x^{-2}$, so by the rule above the relative error is multiplied by $|-2| = 2$, giving 2 %. The sign of the index tells you the direction of the change but not its size: an overestimate of x gives an underestimate of $\dfrac{1}{x^2}$.

14 Exact for every linear f

If $f(x) = mx + c$ then $f'(x) = m$ for every x, and $f(a + h) = m(a + h) + c = (ma + c) + mh = f(a) + f'(a)h$. The estimate and the true value agree exactly, for any h. This matches the previous questions: the error term involves f'', and $f'' \equiv 0$ for a straight line.

15 $h \le 0.057$, roughly

The true value is $1 + 3h + 3h^2 + h^3$ and the linear estimate is $1 + 3h$, so the error is $3h^2 + h^3$. Setting $3h^2 + h^3 = 0.01$ and solving numerically gives $h \approx 0.0573$. At $h = 0.057$ the error is 0.00993; at $h = 0.058$ it is 0.01028.

Topic 13 · Modelling with the derivative

Focus · easy

1 $2t + 3$

2 7

3 $6t$

4 $t = 0$ and $t = 4$

5 10

6 metres per second

7 5

8 $20 - 2x$

9 25, at $t = 5$

10 4

Practice · medium

1 $v = 3t^2 - 12t + 9$, $a = 6t - 12$

2 $t = 1$ and $t = 3$

3 0

4 200π cm³/s

5 $\dfrac{2}{\pi}$ cm/s

6 $x = 25$

7 20

8 24 cm²/s

9 $\dfrac{1}{4}$

10 9, at $t = 3$

Challenge · hard

1 $t = 3$

2 8

6 $\dfrac{1}{5\pi}$ m/s

3 $\frac{5}{6}$ m/s downwards

4 $\frac{2}{\pi}$ cm/s

5 0.75 m/s

7 $x = 20$

8 -13.5, at $t = 2.5$

9 $\frac{dr}{dt}$ is constant

10 $\frac{17}{3}$

Beyond · very hard

1 28

$v = 3t^2 - 18t + 24 = 3(t - 2)(t - 4)$, so the particle rests at $t = 2$ and $t = 4$. The positions are $s(0) = 0$, $s(2) = 20$, $s(4) = 16$ and $s(5) = 20$. Adding the moduli of the changes: $20 + 4 + 4 = 28$. The displacement is only 20.

2 $\frac{2}{5\pi}$ cm/s

With $r = h$, $V = \frac{1}{3}\pi h^3$, so $\frac{dV}{dh} = \pi h^2$ and $\frac{dV}{dt} = \pi h^2 \frac{dh}{dt}$. At $h = 5$: $10 = 25\pi \frac{dh}{dt}$, giving $\frac{dh}{dt} = \frac{2}{5\pi} \approx 0.127$ cm/s. The level slows as the cone widens, since the rate goes as $\frac{1}{h^2}$.

3 $\frac{40}{3} \approx 13.3$ m/s

With x measured along the shore from the foot of the perpendicular, $x = 100 \tan\theta$. Then $\frac{dx}{dt} = 100 \sec^2\theta \cdot \frac{d\theta}{dt}$. At $\theta = 30°$, $\sec^2\theta = \frac{4}{3}$, so $\frac{dx}{dt} = 100 \times \frac{4}{3} \times 0.1 = \frac{40}{3}$ m/s. The spot races away as θ approaches 90°.

4 At $R'(x) = C'(x)$

Profit is $P(x) = R(x) - C(x)$, so $P'(x) = R'(x) - C'(x)$. A stationary point needs $P' = 0$, that is $R' = C'$. For it to be a maximum $P'' < 0$, so $R'' < C''$ — marginal revenue must be falling faster than marginal cost at that output.

5 $x = 25$ at a price of 50

Revenue is $R = px = 100x - 2x^2$, so $R' = 100 - 4x$, zero at $x = 25$, and $R'' = -4 < 0$, a maximum. The price is then $p = 100 - 50 = 50$ and the revenue 1250. Revenue is greatest at half the output that would drive the price to zero.

6 $4\sqrt{3} \approx 6.93$ m/s

With x the horizontal distance and L the string, $x^2 + 100^2 = L^2$. When $L = 200$, $x = \sqrt{30000} = 100\sqrt{3}$. Differentiating, $x\frac{dx}{dt} = L\frac{dL}{dt}$, so $\frac{dL}{dt} = \frac{100\sqrt{3} \times 8}{200} = 4\sqrt{3}$ m/s. It is less than 8, as it must be — only part of the motion is along the string.

7 $a_{acc} = 2a$

Differentiating twice gives $v = 2at + b$ and then $\frac{d^2s}{dt^2} = 2a$, free of t. So a quadratic displacement is exactly the motion of constant acceleration, with b the initial velocity and c the initial displacement. This is where the formula $s = ut + \frac{1}{2}at^2$ comes from.

8 $s = t^3 + t + 2$

Reversing the derivative, $v = 3t^2 + C$, and $v(0) = 1$ gives $C = 1$, so $v = 3t^2 + 1$. Reversing again, $s = t^3 + t + K$, and $s(0) = 2$ gives $K = 2$. Each reversal needs its own initial condition — the subject of Topic 4.

9 $10\sqrt{3} \approx 17.3$ cm²/s

The area is $A = \frac{\sqrt{3}}{4}s^2$, so $\frac{dA}{dt} = \frac{\sqrt{3}}{2}s\frac{ds}{dt}$. At $s = 10$ with $\frac{ds}{dt} = 2$ this is $\frac{\sqrt{3}}{2} \times 10 \times 2 = 10\sqrt{3}$ cm²/s.

10 100 km/h, constant

After t hours the distances are $60t$ and $80t$, so the separation is $d = \sqrt{(60t)^2 + (80t)^2} = 100t$. Hence $\frac{dd}{dt} = 100$ km/h at every instant. Because both motions are steady and perpendicular, the separation grows linearly — a 3–4–5 triangle scaled by $20t$.

11 15 cm³/s

With edge a, $S = 6a^2$ gives $12 = 12a\frac{da}{dt}$, so $\frac{da}{dt} = \frac{1}{a} = \frac{1}{5}$ cm/s. Then $V = a^3$ gives $\frac{dV}{dt} = 3a^2\frac{da}{dt} = 3 \times 25 \times \frac{1}{5} = 15$ cm³/s.

12 $\frac{n-1}{n}$

Here $v = nt^{n-1}$ and $a = n(n-1)t^{n-2}$. So $as = n(n-1)t^{2n-2}$ and $v^2 = n^2t^{2n-2}$. The powers of t cancel exactly, leaving $\frac{n-1}{n}$. For $n = 2$ — constant acceleration — it is $\frac{1}{2}$.

13 $s = \frac{(1+t)^3 - 1}{3}$; distance $\frac{26}{3}$

Reversing the derivative, $s = \frac{(1+t)^3}{3} + C$, and $s(0) = 0$ gives $C = -\frac{1}{3}$. At $t = 2$, $s = \frac{27-1}{3} = \frac{26}{3} \approx 8.67$. Since $v > 0$ throughout, distance and displacement agree.

14 $x = 10$, where the marginal cost is 100

The marginal cost is $C'(x) = x^2 - 20x + 200$. Differentiating again, $C''(x) = 2x - 20$, zero at $x = 10$, and $C''' = 2 > 0$, so it is a minimum of C'. The value is $100 - 200 + 200 = 100$. Note this is where $C'' = 0$ — the point of inflection of the cost curve.

15 0.8 cm²/s

$V = \frac{4}{3}\pi r^3$ gives $\frac{dV}{dt} = 4\pi r^2\frac{dr}{dt}$, so $-2 = 100\pi\frac{dr}{dt}$ and $\frac{dr}{dt} = -\frac{1}{50\pi}$. Then $S = 4\pi r^2$ gives $\frac{dS}{dt} = 8\pi r\frac{dr}{dt} = 8\pi \times 5 \times (-\frac{1}{50\pi}) = -0.8$ cm²/s. The π cancels, which is a useful check.

Topic 14 · Revision · Control work 3 — the derivative in use

Focus · easy

1 $4x^3$

2 $6(2x+1)^2$

3 10

4 5

5 1.02

6 $x = 4$

7 $-\frac{6}{x^4}$

8 100

9 $24t$

10 3.0167

Practice · medium

1 $x = 2$ and $x = -2$

2 Maximum at $x = -2$, minimum at $x = 2$

3 $y = 12x - 16$

4 24π cm²/s

5 3.0111

6 36 cm²

7 $\frac{4}{5}$

8 $t = 1$

9 1.5 %

10 $x = 2$ and $x = -2$

Challenge · hard

1 $y = 4x - 6$

2 $r = 3$ cm

3 0.4849

4 $t = 0$ and $t = 2$

5 0.75 m/s

6 Maximum at $x = 0$, minimum at $x = \frac{8}{3}$

7 $\frac{32\sqrt{6}}{9} \approx 8.71$

8 8.06

9 1 cm/s

10 $\frac{dy}{dx} = \frac{3}{(x+2)^2} > 0$

Beyond · very hard

1 41

$v = 4t^3 - 16t = 4t(t^2 - 4)$, so the particle rests at $t = 0$ and $t = 2$. The positions are $s(0) = 0$, $s(2) = 16 - 32 = -16$ and $s(3) = 81 - 72 = 9$. The distances are 16 and 25, totalling 41, while the displacement is only 9.

2 $12\sqrt{3} \approx 20.8$

With corners at $\pm x$ the area is $A = 2x(9 - x^2) = 18x - 2x^3$, for $0 < x < 3$. Then $A' = 18 - 6x^2 = 0$ gives $x = \sqrt{3}$, and $A'' = -12x < 0$, a maximum. The area is $2\sqrt{3}(9 - 3) = 12\sqrt{3}$.

3 $\frac{n(n-1)}{2}h^2$

The binomial expansion is $1 + nh + \frac{n(n-1)}{2}h^2 + \ldots$, and the linear estimate keeps only $1 + nh$. The leading term of what is left is $\frac{n(n-1)}{2}h^2$. For $n = 3$ and $h = 0.057$ this gives $3h^2 \approx 0.0097$, matching Topic 1.

4 0.04 m/min

At depth h the surface width is also h, by similar triangles, so the cross-section area is $\frac{h^2}{2}$ and $V = 5h^2$. Then $\frac{dV}{dt} = 10h\frac{dh}{dt}$, so $0.2 = 10(0.5)\frac{dh}{dt}$ and $\frac{dh}{dt} = 0.04$ m/min.

5 The square, of perimeter $2d\sqrt{2}$

With diameter d, a rectangle inscribed in the circle has sides $d\cos\theta$ and $d\sin\theta$. Its perimeter is $P = 2d(\cos\theta + \sin\theta) = 2d\sqrt{2}\sin(\theta + \frac{\pi}{4})$, which is greatest when $\theta = \frac{\pi}{4}$. Both sides are then $\frac{d}{\sqrt{2}}$, so the rectangle is a square.

6 $\frac{\sqrt{3}}{2} \approx 0.866$

Minimise $d^2 = x^2 + (x^2 - 1)^2$. Writing $u = x^2 \geq 0$ gives $d^2 = u + (u-1)^2 = u^2 - u + 1$, least at $u = \frac{1}{2}$. Then $d^2 = \frac{1}{4} - \frac{1}{2} + 1 = \frac{3}{4}$, so $d = \frac{\sqrt{3}}{2}$. Two points are equally near, at $x = \pm\frac{1}{\sqrt{2}}$.

7 $\frac{2\pi L^3}{9\sqrt{3}}$, at height $\frac{L}{\sqrt{3}}$

With $r^2 + h^2 = L^2$, the volume is $V = \frac{1}{3}\pi(L^2 - h^2)h$. Then $\frac{dV}{dh} = \frac{\pi}{3}(L^2 - 3h^2) = 0$ gives $h = \frac{L}{\sqrt{3}}$ and $r^2 = \frac{2L^2}{3}$, so $V = \frac{\pi}{3} \cdot \frac{2L^2}{3} \cdot \frac{L}{\sqrt{3}} = \frac{2\pi L^3}{9\sqrt{3}}$.

8 0.7391

Take $f(x) = \cos x - x$, so $f'(x) = -\sin x - 1$. From $x_0 = 0.75$: $f = -0.01831$ and $f' = -1.68164$, giving $x_1 = 0.73911$. One more step gives $x_2 = 0.739085$. The value is the Dottie number, the only real solution.

9 Exactly one real root

A stationary inflection means $y' = 3a(x - p)^2$ for some p, since y' is a quadratic with a repeated root. Integrating, $y = a(x - p)^3 + k$. Setting $y = 0$ gives $(x - p)^3 = -\dfrac{k}{a}$, and a real number has exactly one real cube root, so there is exactly one real solution.

10 108 cm³

With base edge x and height h, $x^2 + 4xh = 108$ gives $h = \dfrac{108 - x^2}{4x}$. Then $V = x^2h = \dfrac{108x - x^3}{4}$, so $\dfrac{dV}{dx} = \dfrac{108 - 3x^2}{4} = 0$ at $x = 6$. Then $h = \dfrac{108 - 36}{24} = 3$ and $V = 36 \times 3 = 108$ cm³.

11 About 0.020 s longer

At $L = 1$, $T = 2\pi\sqrt{\dfrac{1}{9.8}} = 2.0071$ s. The relative change in T is half that in L, so $\dfrac{\Delta T}{T} \approx \dfrac{1}{2}(0.02) = 0.01$ and $\Delta T \approx 0.0201$ s. The clock loses about 1 % of its beats.

12 9

Speed is $|v|$. The velocity is a parabola with least value at $t = 2$, where $v = 12 - 24 + 9 = -3$. At the endpoints $v(0) = 9$ and $v(4) = 48 - 48 + 9 = 9$. The candidates for greatest speed are 9, 9 and $|-3| = 3$, so the greatest speed is 9.

13 $\dfrac{dV}{dr} = 4\pi r^2 = S$

$V = \dfrac{4}{3}\pi r^3$, so $\dfrac{dV}{dr} = 4\pi r^2$, which is the surface area. Geometrically, growing the radius by δr adds a thin shell of volume about $S \cdot \delta r$, so the rate of change of volume per unit radius is the area of the surface being coated. The same holds for a circle: $\dfrac{dA}{dr} = 2\pi r$, the circumference.

14 $a = 1$, $b = -3$

$\dfrac{dy}{dx} = 3ax^2 + b$, and a stationary point at $x = 1$ needs $3a + b = 0$, so $b = -3a$. Passing through $(1, -2)$ gives $a + b = -2$, that is $a - 3a = -2a = -2$, so $a = 1$ and $b = -3$. The curve is $y = x^3 - 3x$.

15 $2\sqrt{74} \approx 17.2$ m

With the point x from the shorter post, $L = \sqrt{36 + x^2} + \sqrt{64 + (10 - x)^2}$. Setting $\dfrac{dL}{dx} = 0$ gives $\dfrac{x}{\sqrt{36 + x^2}} = \dfrac{10 - x}{\sqrt{64 + (10 - x)^2}}$ — the two ropes make equal angles with the ground — so $\dfrac{x}{6} = \dfrac{10 - x}{8}$ and $x = \dfrac{30}{7}$. Reflecting one post below ground makes it a straight line of length $\sqrt{(6 + 8)^2 + 10^2} = \sqrt{296} = 2\sqrt{74}$.

Topic 15 · The antiderivative and the indefinite integral

Focus · easy

1 $\dfrac{x^2}{2} + C$

2 $\dfrac{x^3}{3} + C$

3 $3x + C$

6 x^4

7 C

8 $\dfrac{x^3}{3} - \dfrac{x^2}{2} + C$

9 $x^2 + 5$

4 $\frac{x^6}{6} + C$

5 $x^2 + x + C$

10 x^2

Practice · medium

1 $x^3 - 2x^2 + x + C$

2 $\frac{2}{3}x^{3/2} + C$

3 $-\frac{1}{x} + C$

4 $\frac{(x+1)^3}{3} + C$

5 $-\frac{1}{x} + C$

6 $x^3 + 3$

7 $5x - x^2 + C$

8 $3t^2 + 2$

9 $\frac{x^4}{4} + \frac{x^2}{2} + C$

10 $y = x^2 + 1$

Challenge · hard

1 $\frac{x^5}{5} + \frac{2x^3}{3} + x + C$

2 $x + \frac{1}{x} + C$

3 $\frac{2}{3}x^{3/2} + 2\sqrt{x} + C$

4 $\frac{x^3}{3} + x^2 + C$

5 $1 - \frac{1}{x}$

6 t^3

7 $\frac{(2x+3)^5}{10} + C$

8 $2\sqrt{x} + C$

9 $y = x^3 - 3x^2 + 4$

10 $\frac{x^3}{3} - x + C$

Beyond · very hard

1 They differ by a constant

Let $F' = G' = f$ on an interval and put $H = F - G$. Then $H' = 0$ throughout. By the mean value theorem, for any two points $a < b$ there is c between them with $H(b) - H(a) = H'(c)(b - a) = 0$, so H takes the same value everywhere. Hence $F = G + C$. The interval matters — see question 3.

2 Correct for every x

Differentiate the proposed answer. From Quarter I, $\frac{d}{dx}(x|x|) = 2|x|$, so $\frac{d}{dx}\frac{x|x|}{2} = |x|$, which is the integrand.

Writing it as two cases also works: $\frac{x^2}{2}$ for $x \geq 0$ and $-\frac{x^2}{2}$ for $x < 0$, which join smoothly at the origin.

3 $f(x) = 1 - \frac{1}{x}$ for $x > 0$ and $f(x) = -1 - \frac{1}{x}$ for $x < 0$

The antiderivative is $-\frac{1}{x} + C$, but the domain is split at 0, so each branch carries its own constant. On $x > 0$: $-1 + C_1 = 0$ gives $C_1 = 1$. On $x < 0$: $1 + C_2 = 0$ gives $C_2 = -1$. "Antiderivatives differ by a constant" holds on an interval, and this domain is not one.

4 At $n = -1$ the answer is $\ln|1 + x| + C$

Differentiating $\frac{(1+x)^{n+1}}{n+1}$ gives $(1+x)^n$ by the chain rule, since the inner derivative is 1. At $n = -1$ the formula divides by zero and fails; the integrand is then $\frac{1}{1+x}$, whose antiderivative is $\ln|1 + x| + C$.

5 $y = x^3 + x + 2$

Integrating once, $\frac{dy}{dx} = 3x^2 + C_1$, and the condition at $x = 0$ gives $C_1 = 1$. Integrating again, $y = x^3 + x + C_2$, and $y(0) = 2$ gives $C_2 = 2$. Each integration needs its own condition — two in all for a second derivative.

6 $\frac{[f(x)]^2}{2} + C$

Differentiate the right-hand side by the chain rule: $\frac{d}{dx}\frac{[f(x)]^2}{2} = f(x) \cdot f'(x)$, which is the integrand. The pattern generalises to $\int f' \cdot f^n \, dx = \frac{f^{n+1}}{n+1} + C$ — recognising it saves a substitution.

7 $y = x^2 + C$; through $(2, 1)$ it is $y = x^2 - 3$

The condition is $\frac{dy}{dx} = 2x$, so $y = x^2 + C$ — a stack of parallel parabolas. Through $(2, 1)$: $1 = 4 + C$ gives $C = -3$. Exactly one member passes through any given point, which is what the constant is for.

8 20 m, at $t = 2$ s

Integrating, $v = -10t + 20$ using $v(0) = 20$, and $s = -5t^2 + 20t$ using $s(0) = 0$. The greatest height is where $v = 0$, at $t = 2$, giving $s = -20 + 40 = 20$ m. It returns to the ground at $t = 4$, by the symmetry of the parabola.

9 It divides by zero; the answer is $\ln|x| + C$

At $n = -1$ the formula would give $\frac{x^0}{0}$, which is undefined. The integrand is $\frac{1}{x}$, and the function whose derivative is $\frac{1}{x}$ is $\ln|x|$, so $\int \frac{dx}{x} = \ln|x| + C$. The modulus matters: it makes the result valid for negative x as well.

10 $\frac{x^2}{2} - x + 2\ln|x + 1| + C$

The fraction is improper, so divide first: $x^2 + 1 = (x + 1)(x - 1) + 2$, giving $\frac{x^2+1}{x+1} = x - 1 + \frac{2}{x+1}$.

Integrating term by term gives $\frac{x^2}{2} - x + 2\ln|x + 1| + C$. Dividing before integrating is the standard first move on any improper algebraic fraction.

11 They coincide

If $y_1' = y_2'$ on an interval then $y_1 - y_2$ has zero derivative, so it is a constant k. At $x = 1$ both equal 3, so $k = 0$ and $y_1 \equiv y_2$. Two members of the same family either coincide or never meet — which is why the curves of $y = x^2 + C$ are parallel.

12 $f(x) = 3x + 1$

$f'' = 0$ gives $f' = C_1$, a constant, and then $f = C_1x + C_2$ — a straight line, as expected when the gradient never changes. From $f(0) = 1$, $C_2 = 1$; from $f(1) = 4$, $C_1 + 1 = 4$ so $C_1 = 3$. Here the two conditions are values rather than a value and a gradient, but two are still needed.

13 At $t = 4$ s, having travelled 32 m

Integrating with $v(0) = 0$ gives $v = 12t - 3t^2 = 3t(4 - t)$, which is zero again at $t = 4$. Integrating again with $s(0) = 0$ gives $s = 6t^2 - t^3$, so $s(4) = 96 - 64 = 32$ m. The velocity stays positive on $(0, 4)$, so distance and displacement agree.

14 Parallel parabolas, all with the same gradient at each x

Each member is the same parabola shifted vertically by C. At any fixed x every member has gradient x, so their tangents there are parallel. Exactly one member passes through each point of the plane, and no two ever meet — the constant counts the vertical shift and nothing else.

15 Sums carry over; products do not

Differentiating, $(F + G)' = F' + G' = f + g$, so the sum rule carries straight across. But $(FG)' = F'G + FG' = fG + Fg$, which is not fg in general. Take $f = g = 1$, so $F = G = x$: then $FG = x^2$ has derivative $2x$, not 1. This is exactly why integration has no product rule.

Topic 16 · The table of integrals and the rules of integration

Focus · easy

1 $\dfrac{x^5}{5} + C$

2 $\ln|x| + C$

3 $e^x + C$

4 $-\cos x + C$

5 $\sin x + C$

6 $\dfrac{(2x + 1)^4}{8} + C$

7 $\dfrac{e^{2x}}{2} + C$

8 $\ln|x + 1| + C$

9 $\tan x + C$

10 $\dfrac{5x^3}{3} + C$

Practice · medium

1 $\dfrac{(3x + 2)^6}{18} + C$

2 $-\dfrac{\cos 3x}{3} + C$

3 $\dfrac{\sin(2x + 1)}{2} + C$

4 $-e^{-x} + C$

5 $\dfrac{1}{2}\ln|2x + 3| + C$

6 $\dfrac{2}{3}(x + 4)^{3/2} + C$

7 $\dfrac{x^3}{3} + 2\ln|x| + C$

8 $\dfrac{3^x}{\ln 3} + C$

9 $-\dfrac{1}{3(3x - 1)} + C$

10 $-\cos x - \sin x + C$

Challenge · hard

1 $\dfrac{x^3}{3} - \dfrac{x^2}{2} - 2x + C$

2 $\dfrac{x^2}{2} + \ln|x| + C$

3 $\dfrac{x}{2} - \dfrac{\sin 2x}{4} + C$

4 $\dfrac{x}{2} + \dfrac{\sin 2x}{4} + C$

5 $\tan x - x + C$

6 $\dfrac{e^{2x}}{2} + 2e^x + x + C$

7 $\sqrt{2x + 1} + C$

8 $\ln(x^2 + 1) + C$

9 $\dfrac{(x^2 + 1)^4}{8} + C$

10 $\tan x + C$

Beyond · very hard

1 Proved by differentiating

Differentiate the proposed answer by the chain rule: $\frac{d}{dx}\frac{(ax+b)^{n+1}}{a(n+1)} = \frac{(n+1)(ax+b)^n \cdot a}{a(n+1)} = (ax+b)^n$, the integrand. The a in the denominator is exactly what cancels the a the chain rule brings down.

2 $\frac{\sin^2 x}{2} + C, -\frac{\cos^2 x}{2} + C, -\frac{\cos 2x}{4} + C$ — all the same family

Taking $f = \sin x$ gives $\frac{\sin^2 x}{2}$; taking $f = \cos x$ gives $-\frac{\cos^2 x}{2}$; using $\sin x \cos x = \frac{\sin 2x}{2}$ gives $-\frac{\cos 2x}{4}$.

They differ only by constants: $\frac{\sin^2 x}{2} = -\frac{\cos^2 x}{2} + \frac{1}{2}$, and $-\frac{\cos 2x}{4} = \frac{\sin^2 x}{2} - \frac{1}{4}$. All three are correct — the arbitrary constant absorbs the difference.

3 $\frac{1}{2}\ln\left|\frac{x-1}{x+1}\right| + C$

Split into partial fractions: $\frac{1}{x^2-1} = \frac{1}{2}\left(\frac{1}{x-1} - \frac{1}{x+1}\right)$. Integrating gives $\frac{1}{2}(\ln|x-1| - \ln|x+1|)$, which combines into $\frac{1}{2}\ln\left|\frac{x-1}{x+1}\right| + C$.

4 $x - \ln|x+1| + C$

The fraction is improper in the sense that the degrees match, so divide: $\frac{x}{x+1} = 1 - \frac{1}{x+1}$. Integrating term by term gives $x - \ln|x+1| + C$. Never attempt such a fraction without dividing first.

5 $\frac{x^2}{2} - \frac{1}{2}\ln(x^2+1) + C$

Divide: $\frac{x^3}{x^2+1} = x - \frac{x}{x^2+1}$. The first term integrates to $\frac{x^2}{2}$. For the second, the numerator is half the derivative of the denominator, so it gives $\frac{1}{2}\ln(x^2+1)$. No modulus is needed, since $x^2 + 1 > 0$.

6 $-\cos x + \frac{\cos^3 x}{3} + C$

Peel off one factor: $\sin^3 x = \sin x(1 - \cos^2 x)$. With $u = \cos x$ and $du = -\sin x\,dx$ the integral becomes $-\int(1 - u^2)du = -u + \frac{u^3}{3}$, that is $-\cos x + \frac{\cos^3 x}{3} + C$. An odd power always yields to this trick.

7 $\ln(1 + e^x) + C$

The numerator is exactly the derivative of the denominator, so the integral is $\ln|1 + e^x| + C$, and the modulus may be dropped since $1 + e^x > 0$ always.

8 $\ln|f(x)| + C$

Differentiate the right-hand side: by the chain rule $\frac{d}{dx}\ln|f(x)| = \frac{f'(x)}{f(x)}$, which is the integrand. Recognising this pattern on sight is worth more than any substitution — it handles $\tan x$, $\frac{2x}{x^2+1}$ and $\frac{e^x}{1+e^x}$ in one step each.

9 $\frac{(\ln x)^2}{2} + C$

Here $\frac{1}{x}$ is the derivative of $\ln x$, so with $f = \ln x$ the integrand is $f \cdot f'$, whose integral is $\frac{f^2}{2}$ by Topic 4. Hence the answer is $\frac{(\ln x)^2}{2} + C$.

10 $\ln|\ln x| + C$

Write the integrand as $\frac{1/x}{\ln x}$, which is $\frac{f'}{f}$ with $f = \ln x$. By the rule above the integral is $\ln|\ln x| + C$. Compare with the previous question: the same two pieces, one as a product and one as a quotient, give quite different answers.

11 $\frac{(x-1)^2}{2} + C$ for $x \geq 1$ and $-\frac{(x-1)^2}{2} + C$ for $x < 1$

For $x \geq 1$ the integrand is $x - 1$, giving $\frac{(x-1)^2}{2}$; for $x < 1$ it is $1 - x$, giving $-\frac{(x-1)^2}{2}$. Both vanish at $x = 1$, so taking the same constant on each side makes the antiderivative continuous — and differentiable, since both one-sided derivatives there are 0.

12 They sum to $x + C$

Since $\sin^2 x + \cos^2 x = 1$, the two integrands add to 1, whose integral is $x + C$. The Challenge answers were $\frac{x}{2} + \frac{\sin 2x}{4}$ and $\frac{x}{2} - \frac{\sin 2x}{4}$; adding them, the $\sin 2x$ terms cancel and $\frac{x}{2} + \frac{x}{2} = x$. A quick and complete check on both.

13 $\ln|x^2 + 3x + 1| + C$

The numerator is exactly the derivative of the denominator, so the $\frac{f'}{f}$ rule applies directly and the integral is $\ln|x^2 + 3x + 1| + C$. The modulus is needed here: the quadratic has real roots, so it does take negative values.

14 $y = -\cos x$

Integrating gives $y = -\cos x + C$. At $x = \frac{\pi}{2}$, $\cos x = 0$, so $0 = 0 + C$ and $C = 0$. The curve is $y = -\cos x$, which is the sine curve shifted a quarter period.

15 $e^x + e^{-x} + C$, which is $2\cosh x + C$

Integrating term by term: $\int e^x dx = e^x$ and $\int e^{-x} dx = -e^{-x}$, so the difference integrates to $e^x + e^{-x}$. Since $\cosh x = \frac{e^x + e^{-x}}{2}$, the answer is $2\cosh x + C$ — the integrand being $2\sinh x$, which mirrors $\int \sin$ and $\int \cos$ without the sign changes.

Topic 17 · Revision · Control work 4 — the indefinite integral

Focus · easy

1 $\frac{x^4}{4} + C$

2 $x + C$

3 $\frac{e^{4x}}{4} + C$

4 $\frac{\sin 4x}{4} + C$

5 $\frac{x^2}{2} + 2x + C$

6 $\ln|x + 3| + C$

7 $\frac{(3x-1)^3}{9} + C$

8 $-\cos(x + 1) + C$

9 $2x^2 + 1$

10 $\frac{2^x}{\ln 2} + C$

Practice · medium

1 $\frac{x^3}{3} - \frac{3x^2}{2} + 2x + C$

6 $-\frac{\cos 2x}{2} + C$

2 $\dfrac{(4x+1)^4}{16}+C$

3 $\dfrac{2}{5}x^{5/2}+C$

4 $\dfrac{e^{3x}}{3}+C$

5 $\ln|x-2|+C$

7 $2t^2+t$

8 $x-\ln|x|+C$

9 $\dfrac{\tan 3x}{3}+C$

10 $y=3-\dfrac{1}{x}$

Challenge · hard

1 $\ln|x^2+x+1|+C$

2 $\dfrac{(x+1)^4}{4}+C$

3 $-\dfrac{(1-x)^6}{6}+C$

4 $-\cos x+\tan x+C$

5 $\dfrac{(x^3+1)^3}{9}+C$

6 $-\dfrac{1}{2}\ln|1-2x|+C$

7 $e^x+\dfrac{e^{2x}}{2}+C$

8 t^2+3t+1

9 $\dfrac{x^2}{2}-\dfrac{4}{3}x^{3/2}+x+C$

10 $\dfrac{x}{2}+\dfrac{\sin x}{2}+C$

Beyond · very hard

1 $\dfrac{2}{5}(x+1)^{5/2}-\dfrac{2}{3}(x+1)^{3/2}+C$

With $u=x+1$ we have $x=u-1$ and $dx=du$, so the integral becomes $\int(u-1)\sqrt{u}\,du=\int(u^{3/2}-u^{1/2})du=\dfrac{2}{5}u^{5/2}-\dfrac{2}{3}u^{3/2}+C$. Substituting back gives the answer. Replacing *everything* in terms of u is what makes the substitution work.

2 $x-\ln(1+e^x)+C$

Write $\dfrac{1}{e^x+1}=1-\dfrac{e^x}{e^x+1}$, which is easy to check by combining the right-hand side. The first term integrates to x and the second, being $\dfrac{f'}{f}$, to $\ln(1+e^x)$. Hence the result.

3 $\dfrac{3x}{8}-\dfrac{\sin 2x}{4}+\dfrac{\sin 4x}{32}+C$

Use $\sin^2x=\dfrac{1-\cos 2x}{2}$ twice. First $\sin^4x=\dfrac{1}{4}(1-2\cos 2x+\cos^2 2x)$, then $\cos^2 2x=\dfrac{1+\cos 4x}{2}$, giving $\dfrac{1}{4}\left(\dfrac{3}{2}-2\cos 2x+\dfrac{\cos 4x}{2}\right)$. Integrating term by term gives the stated answer. Even powers always need the double-angle route.

4 Verified

Differentiating, $\dfrac{d}{dx}\ln|\sec x+\tan x|=\dfrac{\sec x\tan x+\sec^2 x}{\sec x+\tan x}=\dfrac{\sec x(\tan x+\sec x)}{\sec x+\tan x}=\sec x$. The bracket cancels exactly, which is the small miracle that makes this the standard answer.

5 54.1 m, at $t=3$ s

Integrating, $v=-9.8t+29.4$ and $s=-4.9t^2+29.4t+10$. The greatest height is where $v=0$, at $t=3$, giving $s=-44.1+88.2+10=54.1$ m.

6 $x - 2\ln|x+1| - \dfrac{2}{x+1} + C$

Put $u = x + 1$, so $x = u - 1$ and $x^2 + 1 = u^2 - 2u + 2$. The integrand becomes $1 - \dfrac{2}{u} + \dfrac{2}{u^2}$, which integrates to $u - 2\ln|u| - \dfrac{2}{u}$. Substituting back and absorbing the constant 1 gives the answer.

7 F is even

Consider $G(x) = F(-x)$. Then $G'(x) = -F'(-x) = -f(-x) = f(x)$, since f is odd. So G and F have the same derivative and differ by a constant; at $x = 0$ both are 0, so the constant is zero and $F(-x) = F(x)$. For example $f = x$ is odd and $F = \dfrac{x^2}{2}$ is even.

8 $\dfrac{1}{2}\arctan\dfrac{x}{2} + C$

Use the standard form $\int \dfrac{dx}{x^2 + a^2} = \dfrac{1}{a}\arctan\dfrac{x}{a} + C$ with $a = 2$. Checking: $\dfrac{d}{dx}\dfrac{1}{2}\arctan\dfrac{x}{2} = \dfrac{1}{2} \cdot \dfrac{1/2}{1 + x^2/4} = \dfrac{1}{4 + x^2}$, as required.

9 Verified

Differentiate by the product rule: $\dfrac{d}{dx}[e^x(x-1)] = e^x(x-1) + e^x \cdot 1 = e^x(x - 1 + 1) = x\,e^x$, the integrand. Finding this answer rather than checking it needs integration by parts, which is Quarter III.

10 $y = x - \sin x$

Integrating once, $\dfrac{dy}{dx} = -\cos x + C_1$, and the condition at $x = 0$ gives $-1 + C_1 = 0$, so $C_1 = 1$. Integrating again, $y = -\sin x + x + C_2$, and $y(0) = 0$ gives $C_2 = 0$. The curve $y = x - \sin x$ is the cycloid's x-component, and it is increasing everywhere.

11 $\arctan x + \dfrac{1}{2}\ln(1 + x^2) + C$

Split the fraction: $\dfrac{1}{1 + x^2} + \dfrac{x}{1 + x^2}$. The first is the standard arctangent form. In the second the numerator is half the derivative of the denominator, giving $\dfrac{1}{2}\ln(1 + x^2)$. Splitting a sum in the numerator is always allowed; splitting one in the denominator never is.

12 $\dfrac{(x^3 + 5)^5}{15} + C$

Differentiating $\dfrac{f^{n+1}}{n+1}$ by the chain rule gives $f^n f'$, the integrand. For the example, $f = x^3 + 5$ has $f' = 3x^2$, so $x^2 = \dfrac{f'}{3}$ and the integral is $\dfrac{1}{3} \cdot \dfrac{(x^3 + 5)^5}{5} = \dfrac{(x^3 + 5)^5}{15} + C$.

13 36 litres, at $t = 6$ min

The flow stops when $6t - t^2 = t(6 - t) = 0$, that is at $t = 6$ (the root $t = 0$ is the start). Integrating with $V(0) = 0$ gives $V = 3t^2 - \dfrac{t^3}{3}$, so $V(6) = 108 - 72 = 36$ litres. After $t = 6$ the rate turns negative and the tank would drain.

14 $\arcsin\dfrac{x}{2} + C$

Use the standard form $\int \dfrac{dx}{\sqrt{a^2 - x^2}} = \arcsin\dfrac{x}{a} + C$ with $a = 2$. Checking: $\dfrac{d}{dx}\arcsin\dfrac{x}{2} = \dfrac{1/2}{\sqrt{1 - x^2/4}} = \dfrac{1}{\sqrt{4 - x^2}}$. The integral is defined only for $|x| < 2$.

15 No elementary antiderivative exists; it is computed numerically or named

Differentiation is an algorithm: every elementary function has an elementary derivative. Integration is a search, and the elementary functions are not closed under it — Liouville proved that no combination of powers, roots, exponentials, logarithms and trigonometric functions differentiates to e^{x^2}. In practice the integral is evaluated numerically, or given a name: the closely related $\int e^{-x^2} dx$ is the error function, on which the normal distribution rests.